Presentation 4

STAT 390 | Group 2 | Winter 2026

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Part 1: Programmers

Roma & Paige & Charlene

Effect of Patch Order on Solution Quality

Task 2

Goal *Analyzing Effect of Patch Order*

→ batch size is 1 so patches are presented to the model one at a time during training

→ gradient descent trajectory depends on the order in which patches are presented during training

→ Due to the highly non-convex nature of the loss function, different random seeds that control the patch shuffling and order can lead the model to different local optima.

- *Goal: study how much the random seed affects model performance and stability across 5 data splits and 10 seeds, and draw conclusions about how seriously we should take initialization when evaluating and tuning this model.*

Analyses performed

Best Validation Loss: boxplot across 10 seeds per split

Predicted Probabilities: boxplot per test case across 10 seeds

Classification Consistency: % of runs each case is correctly classified

Accuracy Variability: max and min test accuracy per split

Additional: key visualizations, false positive tracking and cross-reference with Optional Task 3

Setup

To study this effect we ran the full training pipeline 50 times total with the 5 saved splits, each across the same random 10 seeds, using a for loop in the sbatch script.

- **The 5 pre-existing no-leakage splits:** data_splits_new_01.npz … data_splits_new_05.npz
- **10 randomized seeds:** 18, 34, 68, 93, 66, 8, 74, 49, 97, 24 (same seeds applied across all 5 splits)
- **Each run saved to its own directory**: named split{n}_seed{n} for easy identification
- **Same outputs saved per run** (i.e. predictions.csv, confusion_matrix, results.txt, attention_analysis.py), but with addition of seed, split, and Best Val. Loss to results.txt for analysis

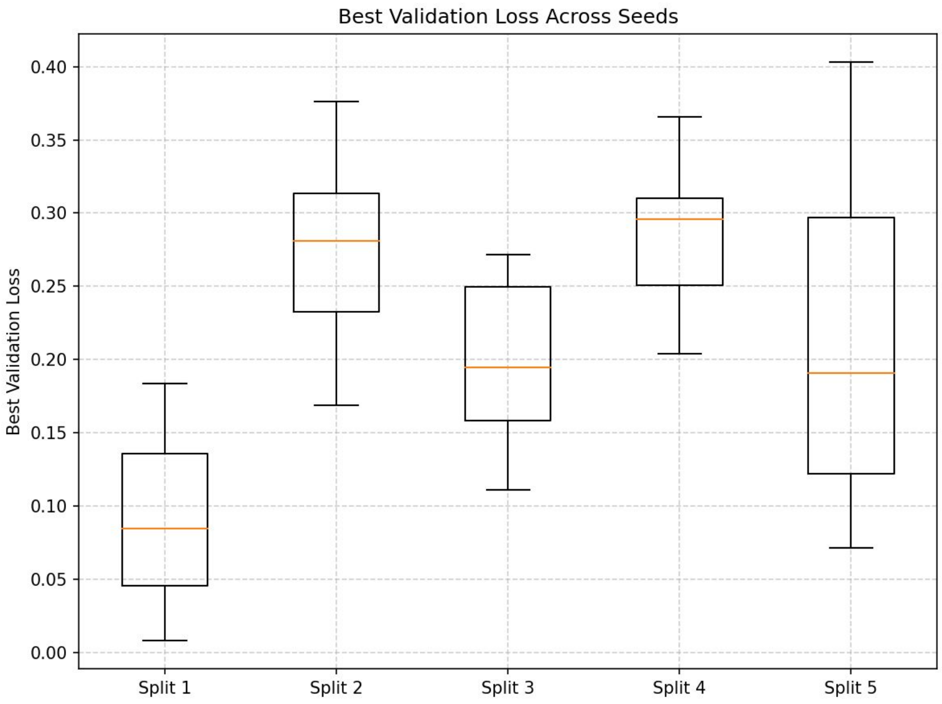
```
cd /projects/e32998/STAT390_Winter2026/Presentation4/STAT390_W

SPLITS=(
    "/projects/e32998/STAT390_Winter2026/Presentation4/STAT390
    "/projects/e32998/STAT390_Winter2026/Presentation4/STAT390
    "/projects/e32998/STAT390_Winter2026/Presentation4/STAT390
    "/projects/e32998/STAT390_Winter2026/Presentation4/STAT390
    "/projects/e32998/STAT390_Winter2026/Presentation4/STAT390
)

SEEDS=(18 34 68 93 66 8 74 49 97 24)

for split_idx in "${!SPLITS[@]}"; do
    split_path="${SPLITS[$split_idx]}"
    split_num=$((split_idx + 1))

    for seed in "${SEEDS[@]}"; do
        echo "Running split${split_num}_seed${seed}..."

        python3 main.py \
            --seed $seed \
            --run_name split${split_num}_seed${seed} \
            --per_slice_cap 500 \
            --max_slices_per_stain 5 \
            --analyze_attention \
            --attention_top_n 8 \
            --load_splits $split_path

        echo "Completed split${split_num}_seed${seed}"
    done
done

echo "All runs completed: $(date)"
```

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/patch_order_splits.sbatch
/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/main.py
/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/utils.py

Analyses

Task 2

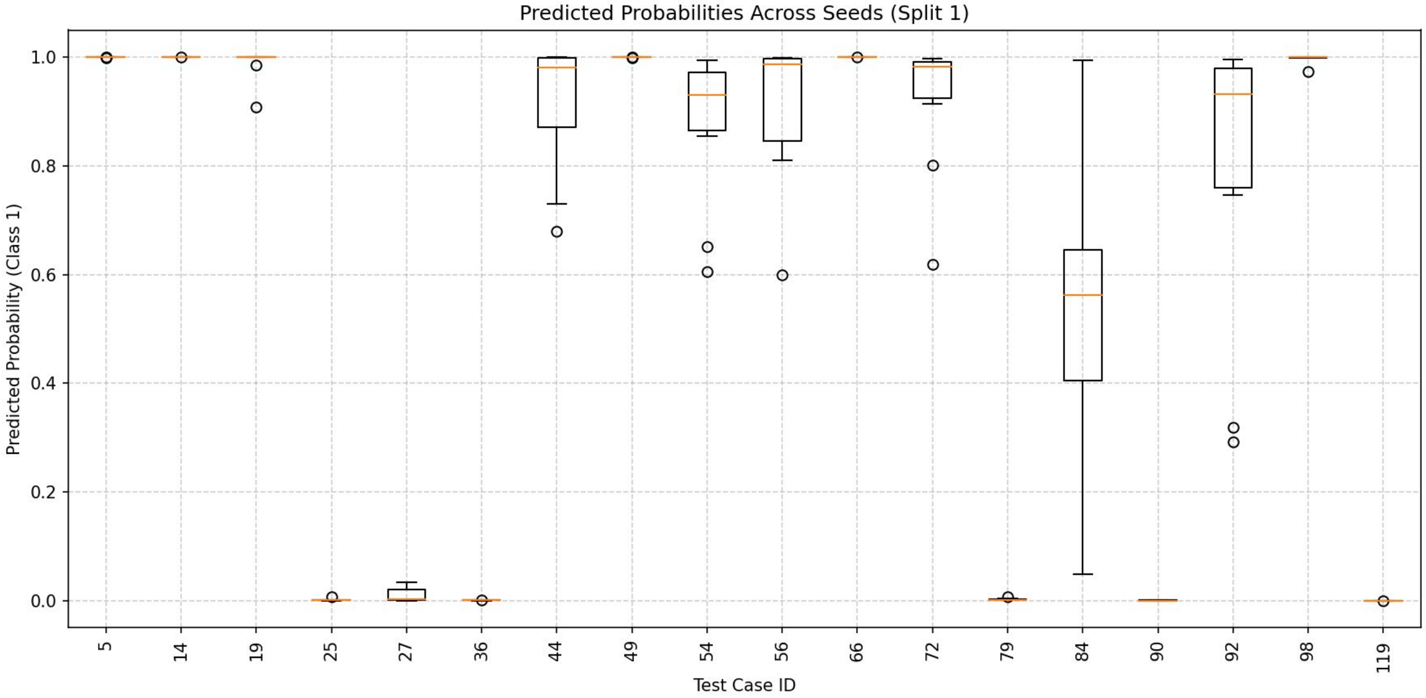
Best Validation Loss



- Validation loss varies across seeds but remains within moderate ranges
    - Split 1: ~0.01 – 0.18 (median ≈ 0.08)
    - Split 2: ~0.17 – 0.37 (median ≈ 0.28)
    - Split 3: ~0.11 – 0.27 (median ≈ 0.19)
    - Split 4: ~0.20 – 0.37 (median ≈ 0.30)
    - Split 5: ~0.07 – 0.40 (median ≈ 0.19)
- Dataset splits influence performance more than seed choice
- Representative seeds (min / median / max) capture the variation

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb

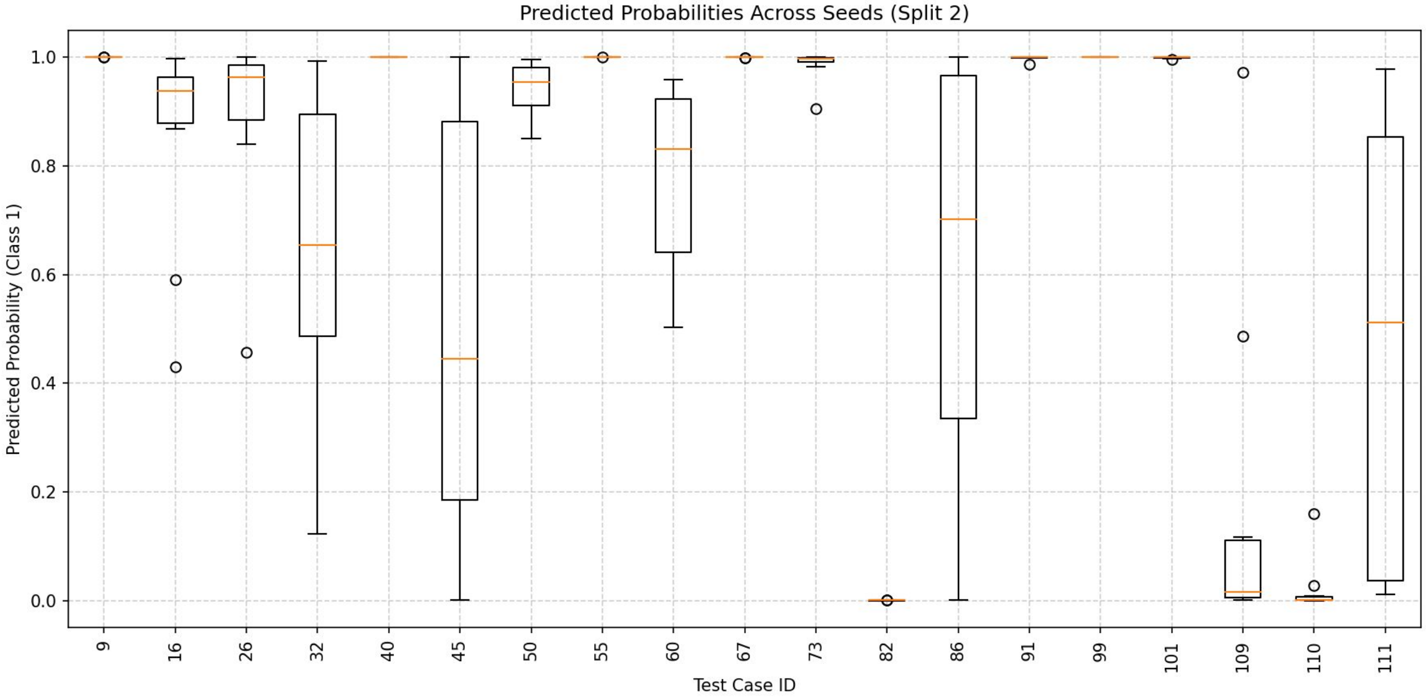
Predicted Probability



- Most test cases show stable predictions across seeds
- Many probabilities consistently near 0 or 1
    - High confidence: 5, 14, 49, 66
    - Low confidence: 25, 36, 79, 119
- Some variability observed in Case 84

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb

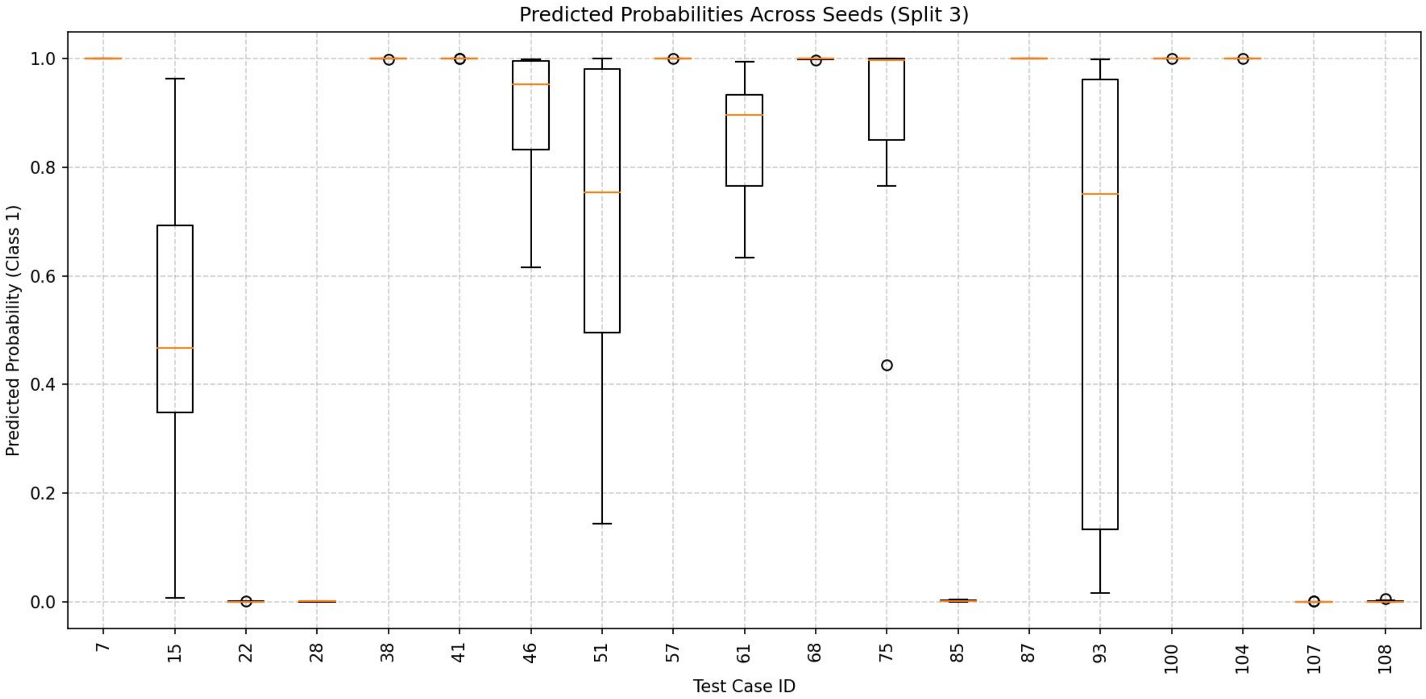
Predicted Probability



- Predictions show more variability across seeds
- Some cases remain highly confident
- High variability examples
  - 32
  - 45
  - 86
- Indicates more difficult or ambiguous samples

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb

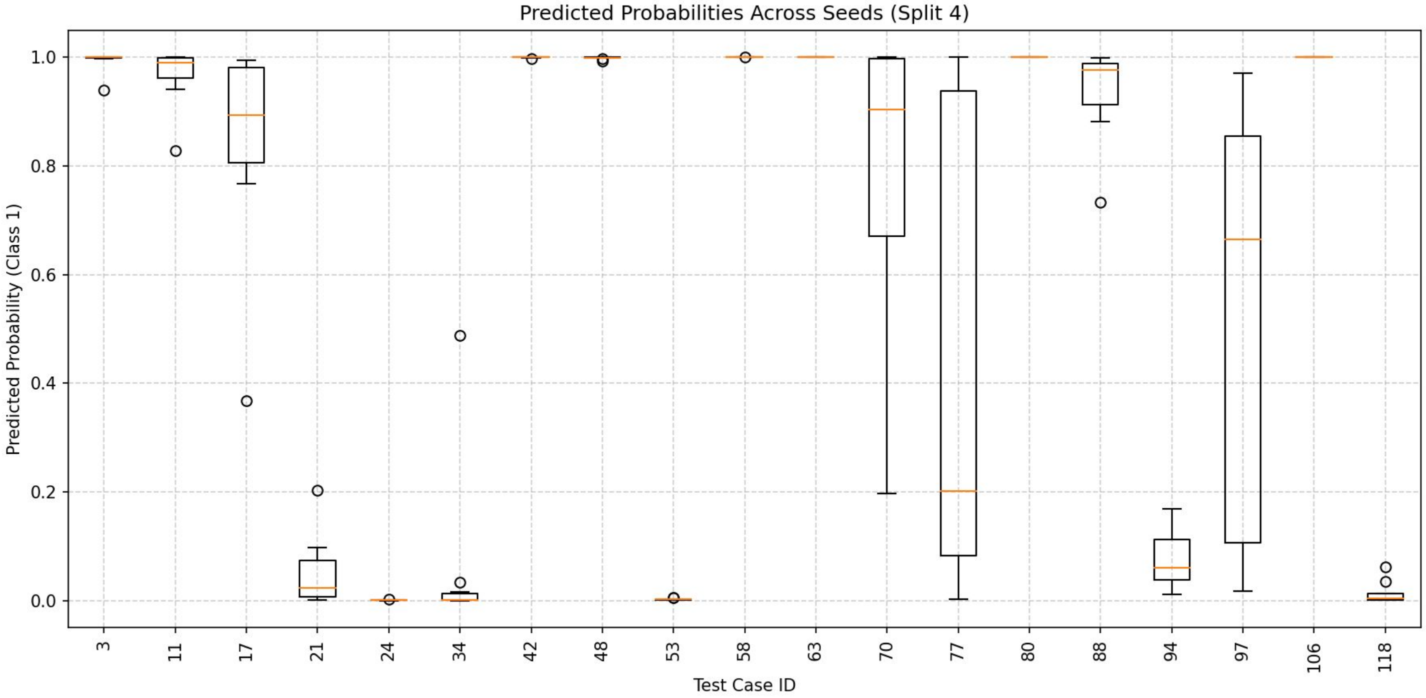
Predicted Probability



- Majority of predictions remain consistent across seeds
- Several cases show large prediction variability
- Notable variable cases:
  - 15
  - 51
  - 93
- Variability likely reflects borderline samples

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb

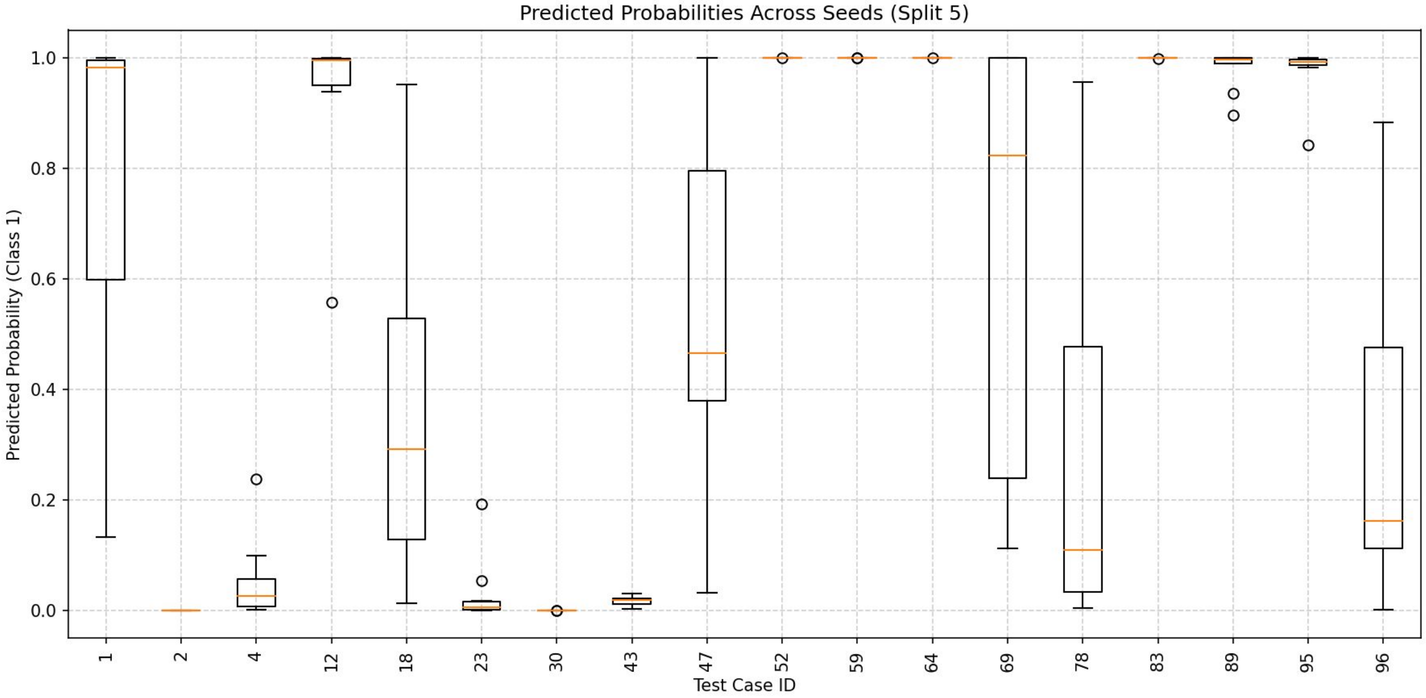
Predicted Probability



- Majority of predictions remain consistent across seeds
- Several cases show large prediction variability
- Notable variable cases:
  - 70
  - 77
  - 97
- Variability likely reflects borderline samples

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb

Predicted Probability



- Most predictions remain consistent across seeds
- Several cases show substantial variability
- Notable variable cases:
  - 1
  - 18
  - 69
  - 96
- Variability likely reflects difficult or borderline samples

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb

Classification Consistency

For each case, we compute the percentage of runs (out of 10) in which the case was classified correctly, only counting the runs belonging to a split where that case appeared in the test set .

This gives us a measure of how stable the cases' classifications are across different random seeds.

- 100% → **independent of seed**: the model correctly classifies this case regardless of patch ordering
- Medium percentage (less than 100%) → **seed sensitive:** the model is inconsistent in classifying the case correctly across the seeds, so patch ordering during training affects results
- 0%→ **consistently wrong:** model fails on this case regardless of seed, suggesting it is a genuinely difficult case

```python
# for each case in this run record if it was correct
for _, row in df.iterrows():
    case_id = row["case_id"]
    correct = bool(row["correct"])
    case_results[case_id].append({
        "split": split,
        "seed": seed,
        "correct": correct,
        "true_label": row["true_label"],
        "predicted_label": row["predicted_label"],
    })

# also find the false positives (benign cases, true label=0,
# and predicted as high grade, predicted label=1) for this run
fp = df[(df["true_label"] == 0) & (df["predicted_label"] == 1)]["case_id"].tolist()
false_positives_per_run[folder] = fp
```

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/paige_classification_analysis.py

Classification Consistency *Results*

- **73.1%** of cases (68/93) are perfectly stable→ classified correctly across all 10 seeds, so regardless of patch ordering
- **25** cases show some instability→ seed affects whether they are classified correctly
- **Only 2** cases are never correct (0%) regardless of seed (case 56, 46) → genuinely hard cases for the model

```
Consistency Summary:
  Total cases: 93
  Cases at 100% consistency: 68 (73.1%)
  Cases below 100% (any instability): 25
  Cases at 0% (never correct): 2

  Of cases with any instability (below 100%):
    Benign cases:     11/25
    High-grade cases: 14/25

  Of cases below 50% consistency:
    Total:            7
    Benign cases:     5/7
    High-grade cases: 2/7
```

classification_consistency

case_id	true_label	total_appearances	correct_count	pct_correct
56	0	10	0	0.0
46	0	10	0	0.0
26	0	10	1	10.0
86	0	10	3	30.0

... *(sorted ascending by pct_correct)*

89	1	10	10	100.0
83	1	10	10	100.0
95	1	10	10	100.0

- Of the 25 unstable cases, instability is relatively evenly split with 11 benign, 14 high-grade
- However looking further at even more unstable cases (below 50%), the breakdown is of more benign cases, being 5 out of the 7

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/paige_classification_analysis.py

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/output_logs2/classification_consistency.csv

Accuracy Variability

Maximum and minimum test accuracy obtained across the 10 runs for each split

Split	Minimum Test Accuracy	Minimum Test Accuracy Seed	Maximum Test Accuracy	Maximum Test Accuracy Seed
1	0.8333	34	0.9444	24, 49, 66, 68, 93
2	0.7895	24, 68, 74	0.8947	18, 49
3	0.7895	24, 74, 93	0.9474	18, 66, 68
4	0.8421	24	1.0000	49, 74
5	0.7222	34	0.9444	8

- Closer look should be taken at seeds 24 and 74 - prevalent seeds in both min and max - potentially unreliable

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/output_logs2

Additional Analysis & Visualizations

Task 2

False Positive Tracking
Across 50 Runs

Also tracked false positives in the test results across all 50 runs to better understand the model's weakness of misclassifying benign cases.

Split	Total FPs Across 10 Seed Runs	Runs with 0 FPs	List of FP Case Ids (Unique)
Split 1	10	0/10	[56]
Split 2	27	0/10	[26, 45, 86, 109, 111]
Split 3	16	0/10	[46, 93]
Split 4	4	6/10	[77]
Split 5	6	5/10	[78, 96]

Key findings from tracking summary:

- 63 total false positives across 50 runs (average of 1.26 per run)
- Only 11/50 runs had zero false positives
- Each split has at least one persistent false positive case that appears regardless of seed

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/paige_classification_analysis.py

Cross-Reference *Optional Task 3*

Optional Task 3 identified Cases 21, 86, 96 as misclassified benign cases using one specific split and seed. Here we cross-referenced with those cases on the seed analysis to see if those misclassifications seem seed dependent.

```
Cross-reference with Optional Task 3 misclassified benign cases:
  Case 21: appeared in 10 runs across 1 split(s) ['split4'], false positive in 0/10 runs (0.0%)
  Case 86: appeared in 10 runs across 1 split(s) ['split2'], false positive in 7/10 runs (70.0%)
  Case 96: appeared in 10 runs across 1 split(s) ['split5'], false positive in 3/10 runs (30.0%)
```

- Case 21: false positive in 0/10 runs; never misclassified regardless of seed in this split
- Case 86: false positive in 7/10 runs; consistently wrong regardless of seed (difficult case)
- Case 96: false positive in 3/10 runs, seed appears to matter here, sometimes correct sometimes not

Note: each case only appeared in one split's test set; results reflect seed sensitivity within that split's fixed train/test composition only.

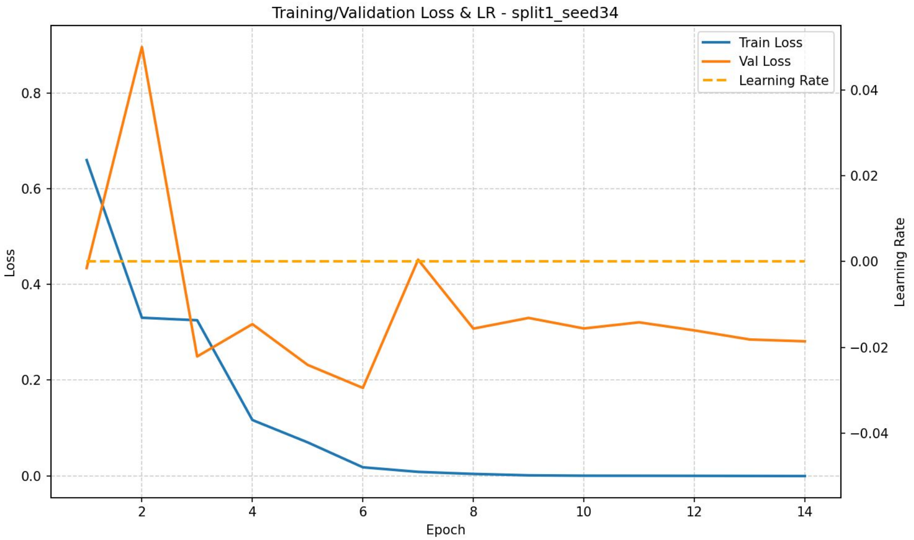
/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Code9_no_leakage/coders_task2/paige_classification_analysis.py

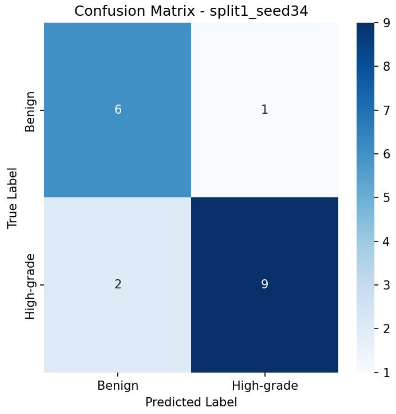
Goal *Analyzing model/output changes per seed*

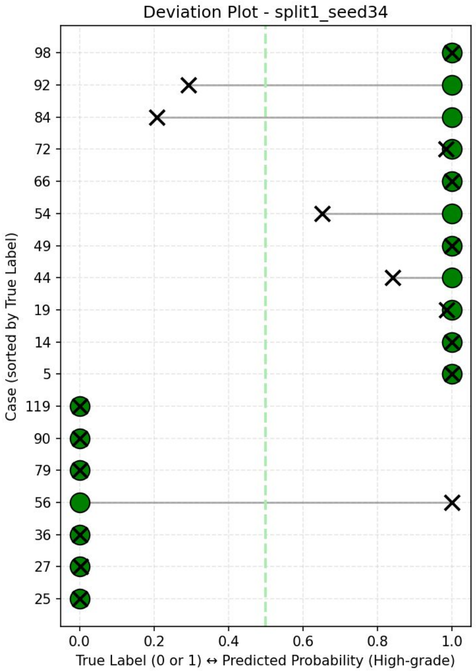
Considering the changes from the seeds and the splits, we want to consider how these aspects differ from one another

- **Training & Validation Loss**
- **Confusion Matrix**
- **Deviation Plot**
- **Visualization of high attention patches for misclassified benign and high grade cases**

Min Split 1 Seed 34







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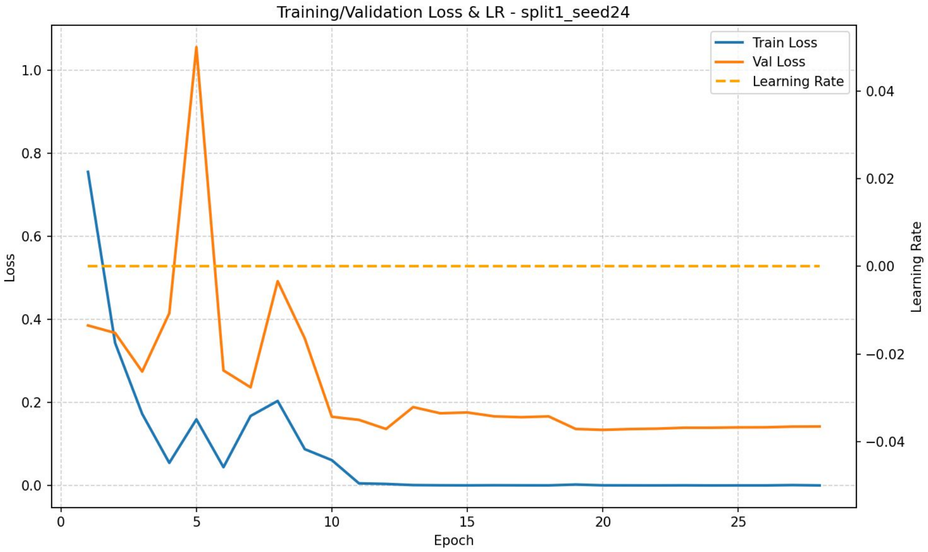
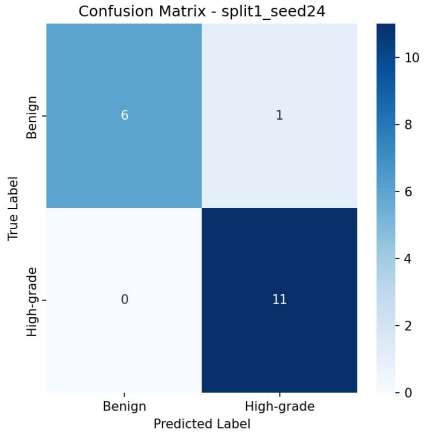
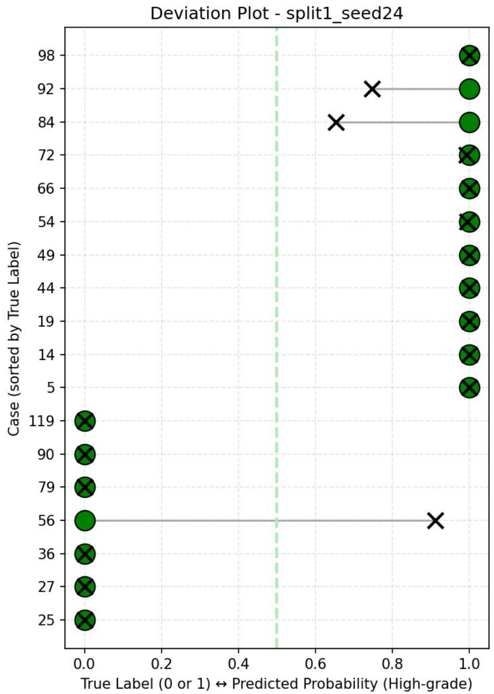
Median Split 1 Seed 24







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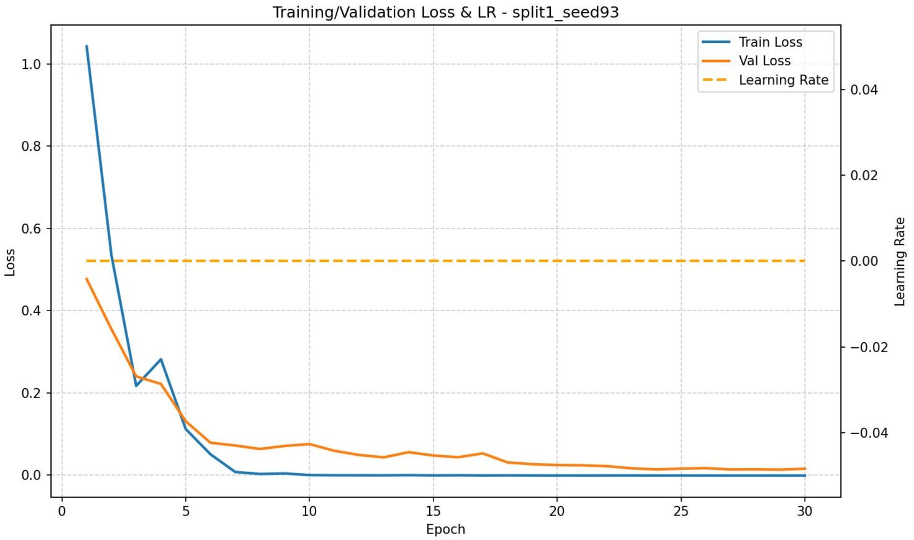
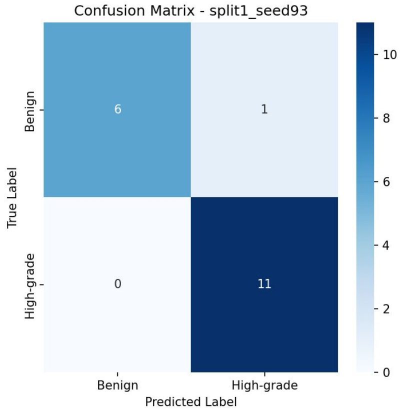
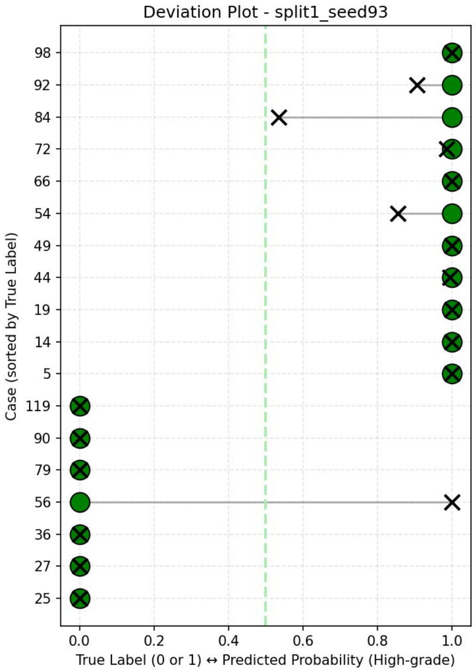
Max Split 1 Seed 93







Split 1 takeaways?

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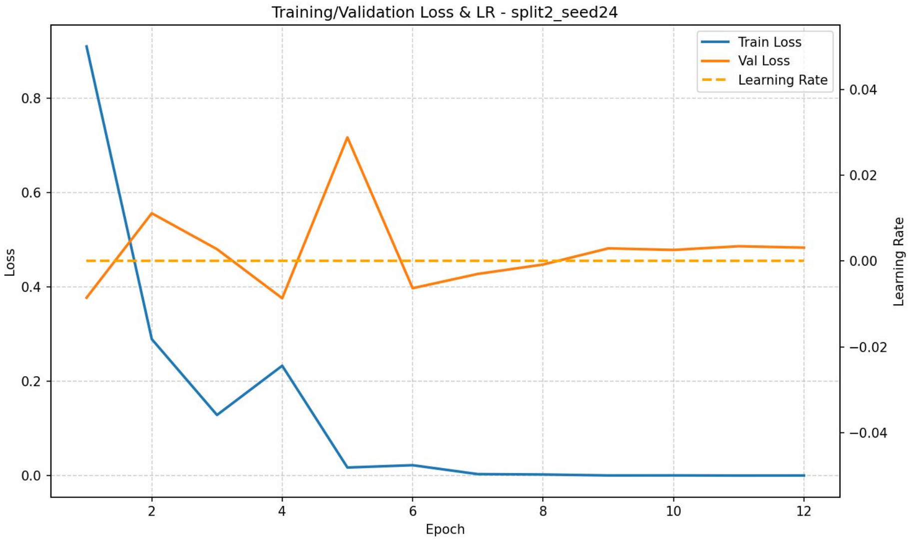
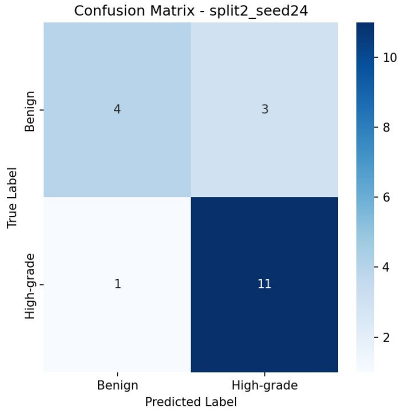
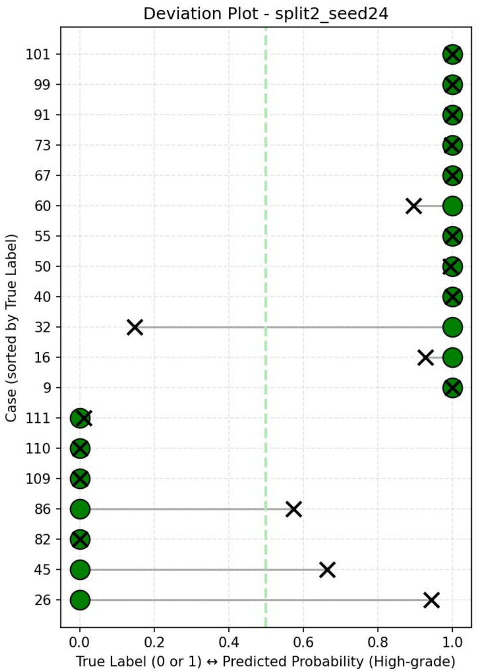
Min Split 2 Seed 24







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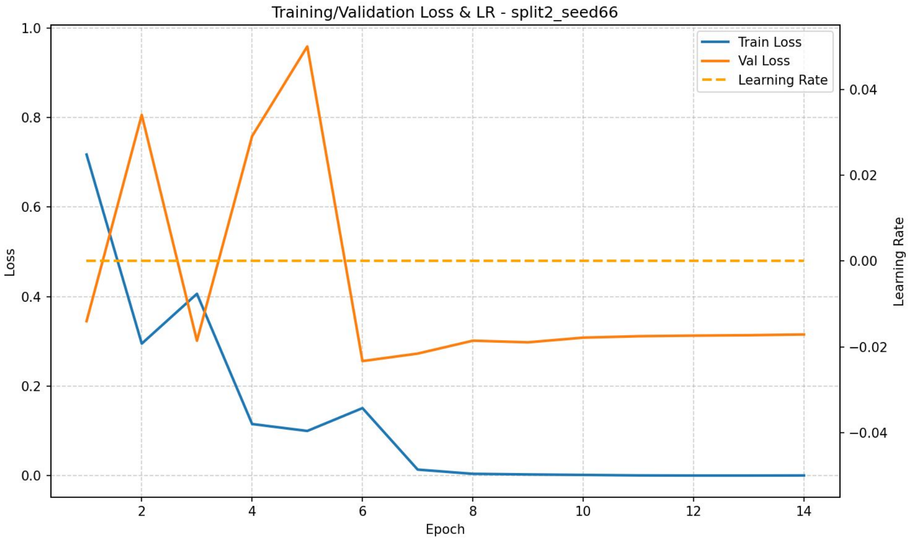
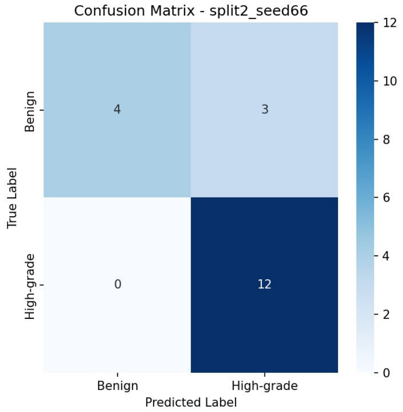
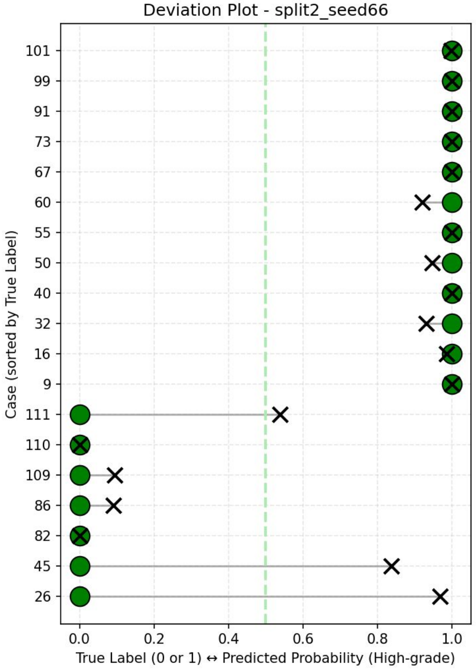
Median Split 2 Seed 66







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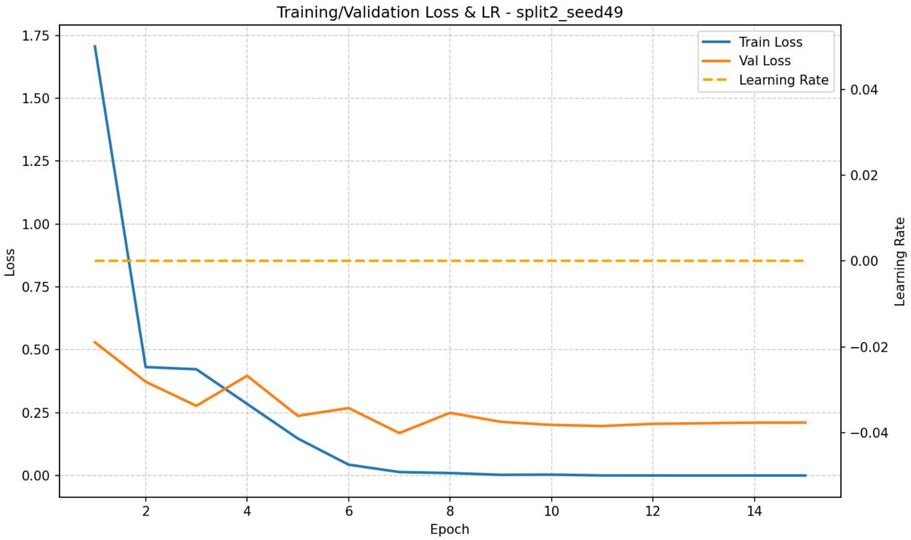
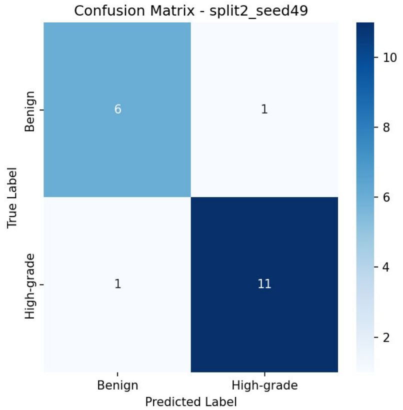
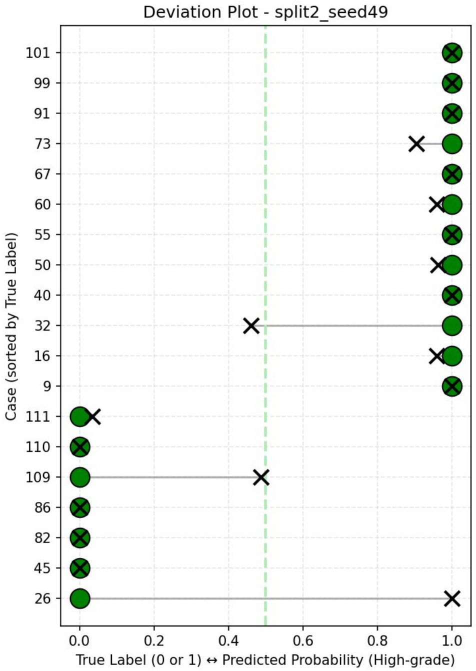
Max Split 2 Seed 49







Split 2 takeaways?

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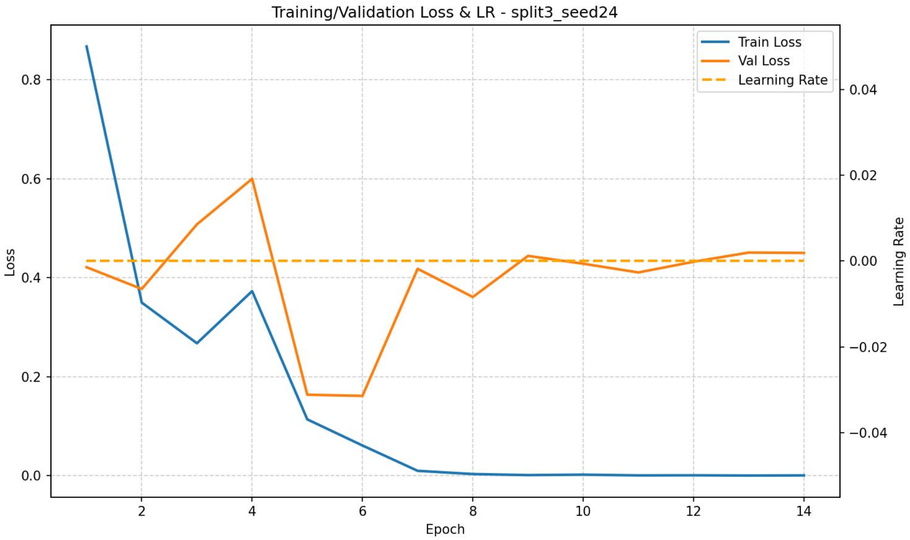
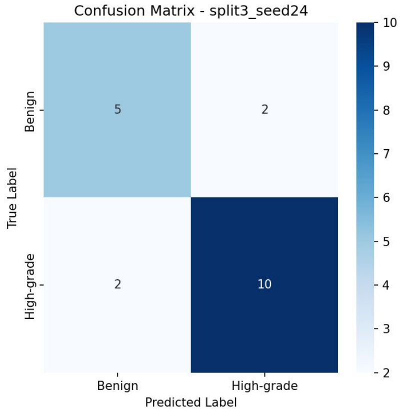
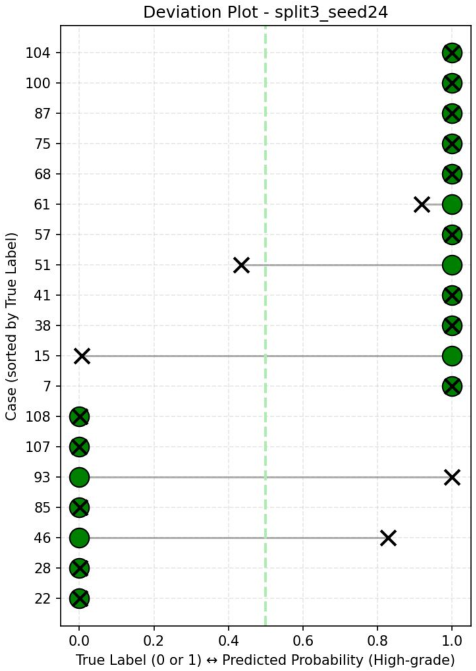
Min Split 3 Seed 24







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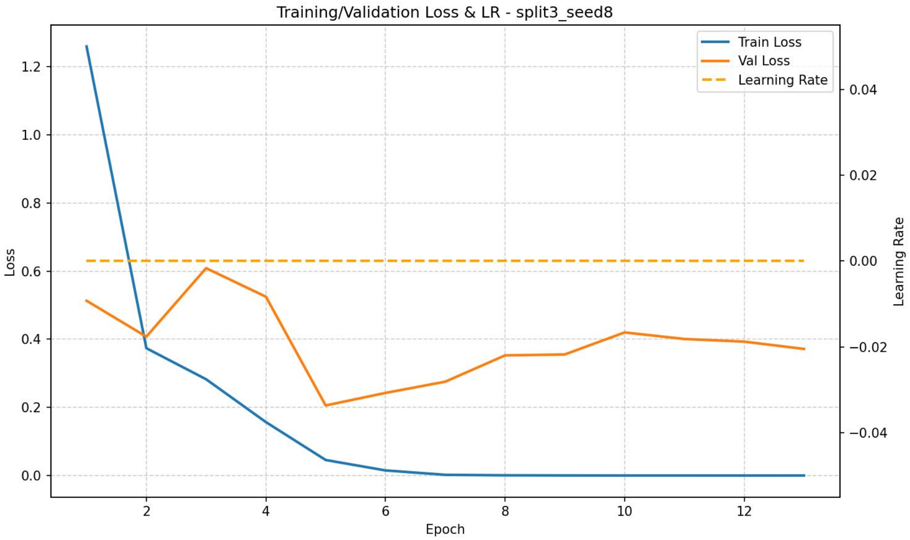
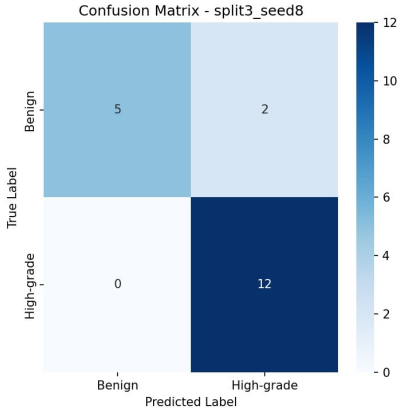
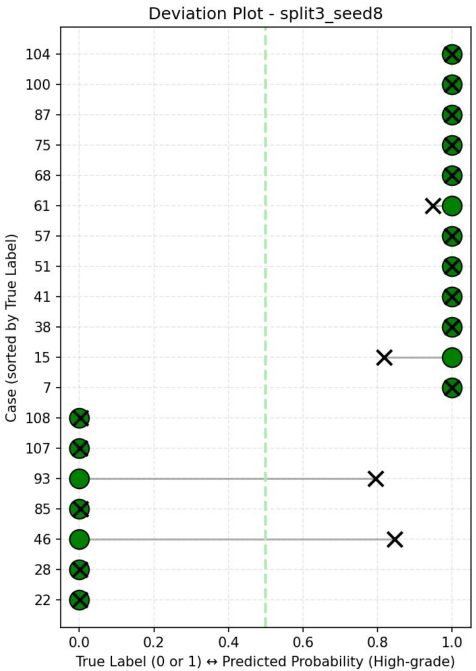
Median Split 3 Seed 8







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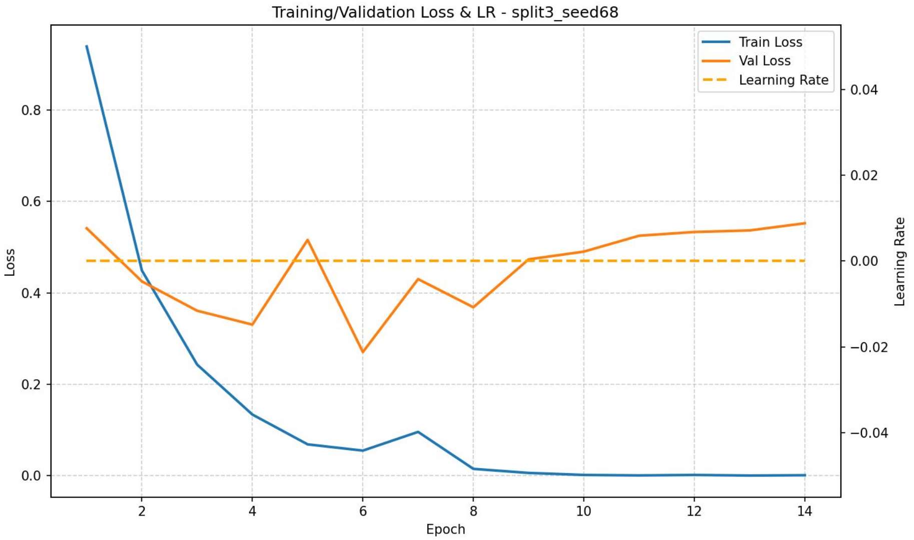
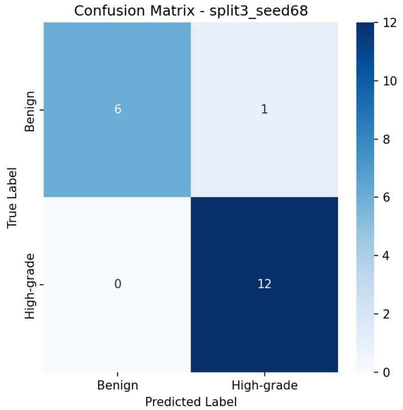
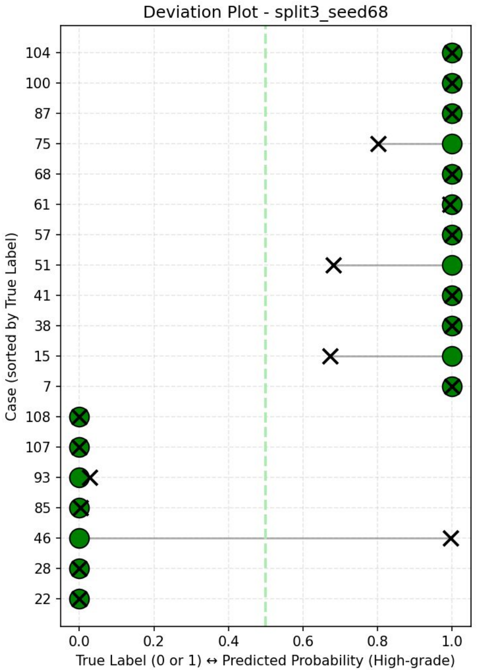
Max Split 3 Seed 68







Split 3 takeaways?

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb
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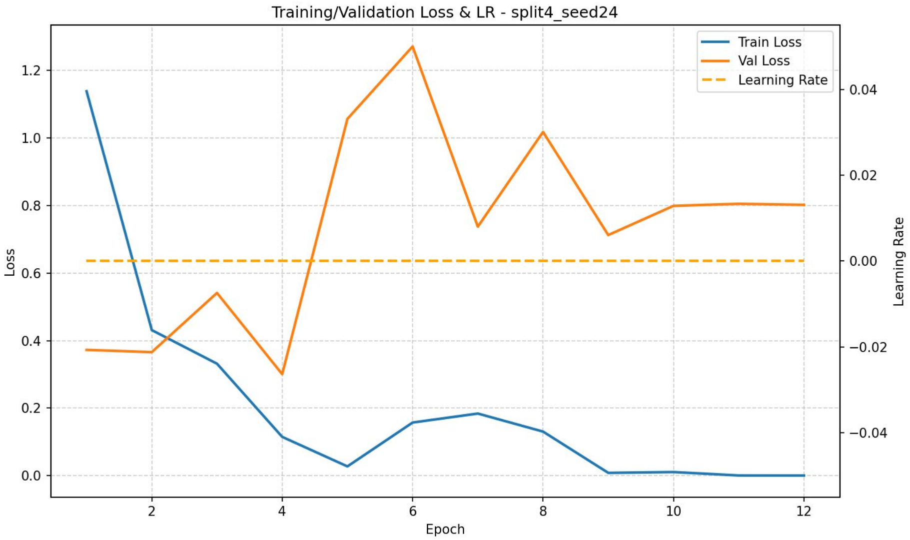
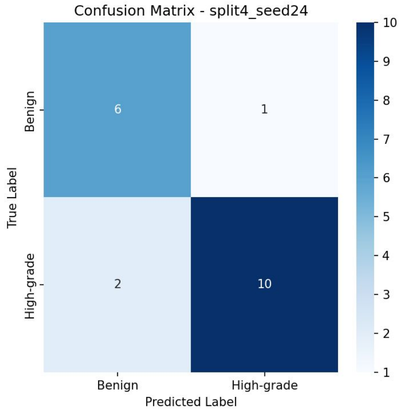
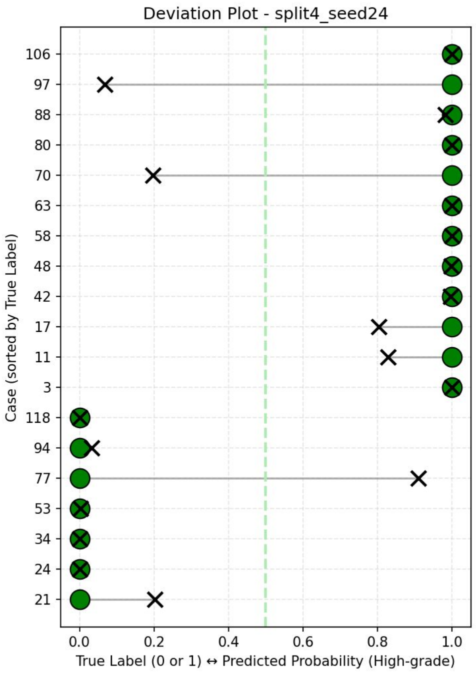
Min Split 4 Seed 24







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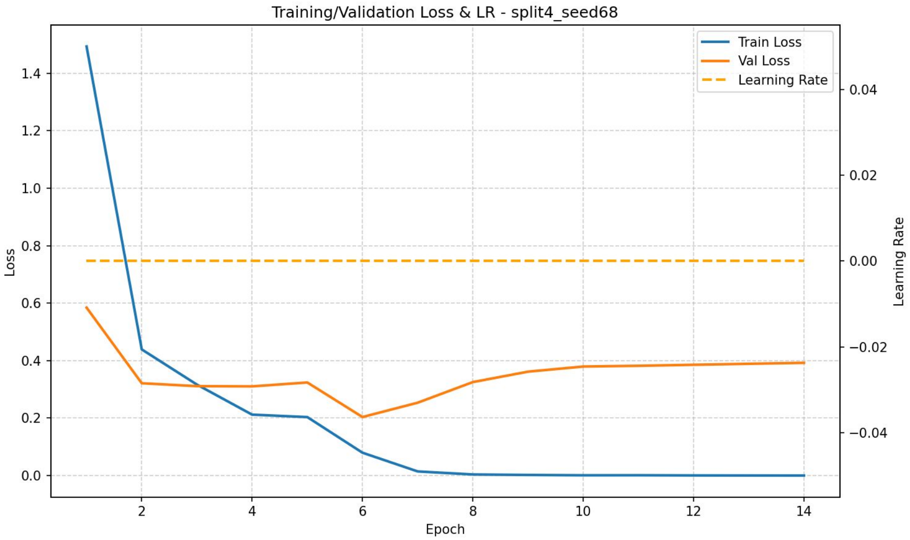
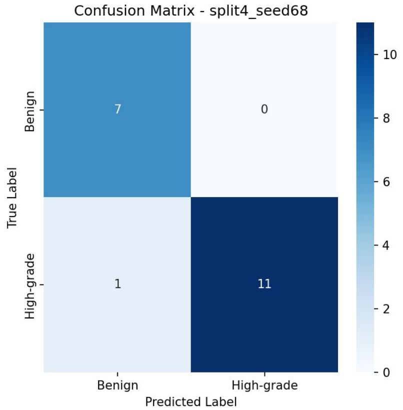
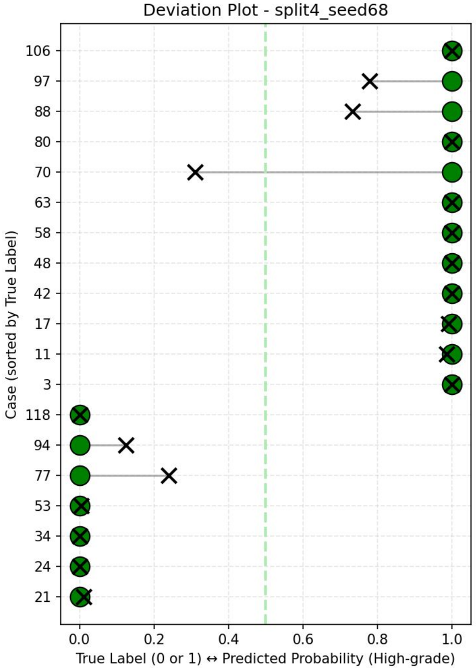
Median Split 4 Seed 68







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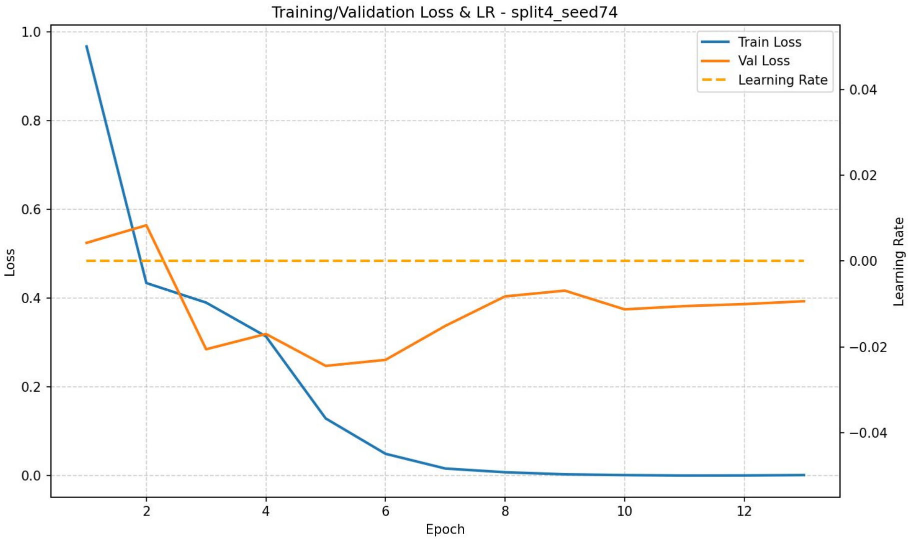
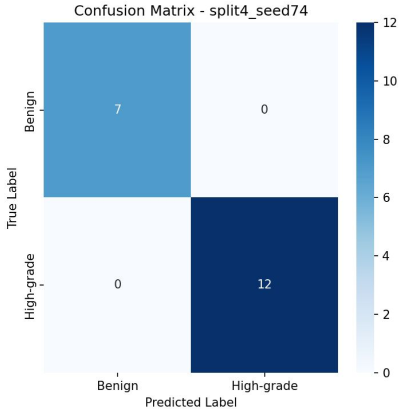
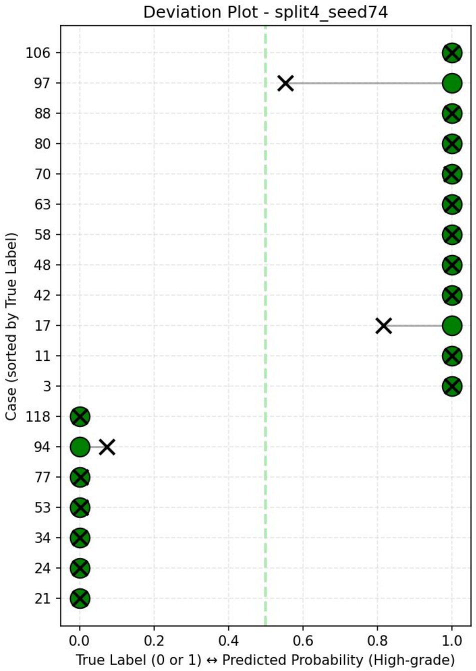
Max Split 4 Seed 74







Split 4 takeaways?

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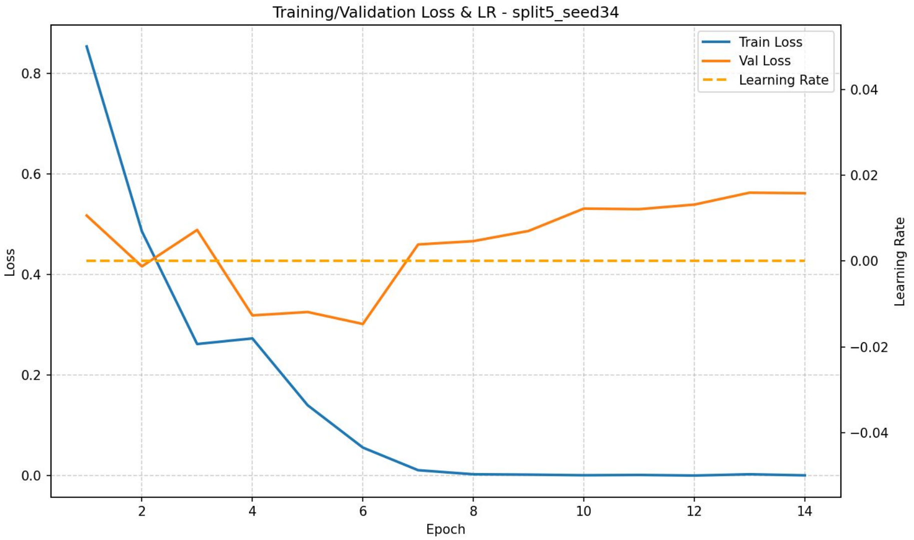
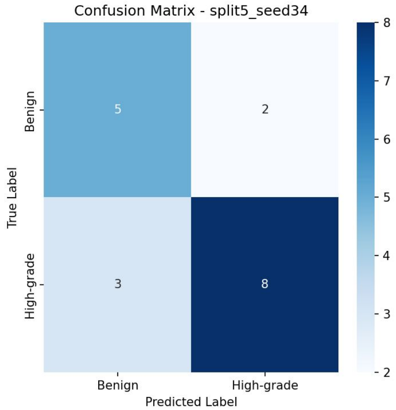
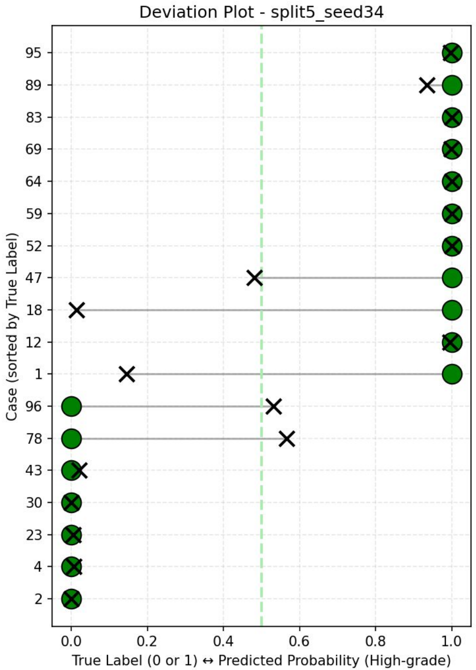
Min Split 5 Seed 34







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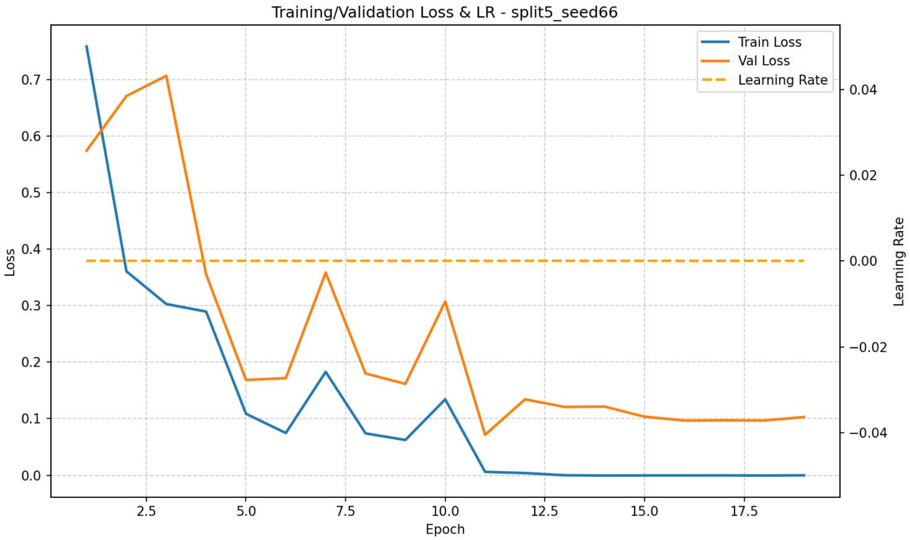
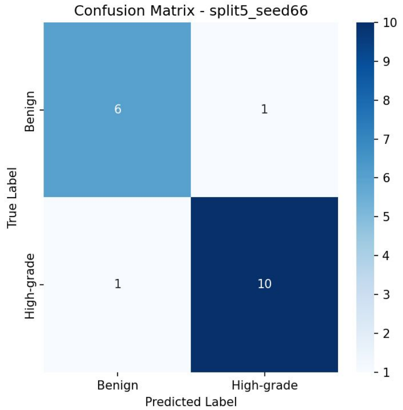
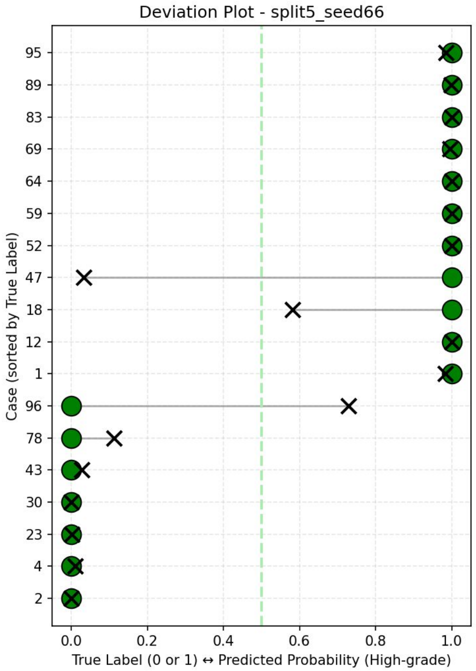
Median Split 5 Seed 66







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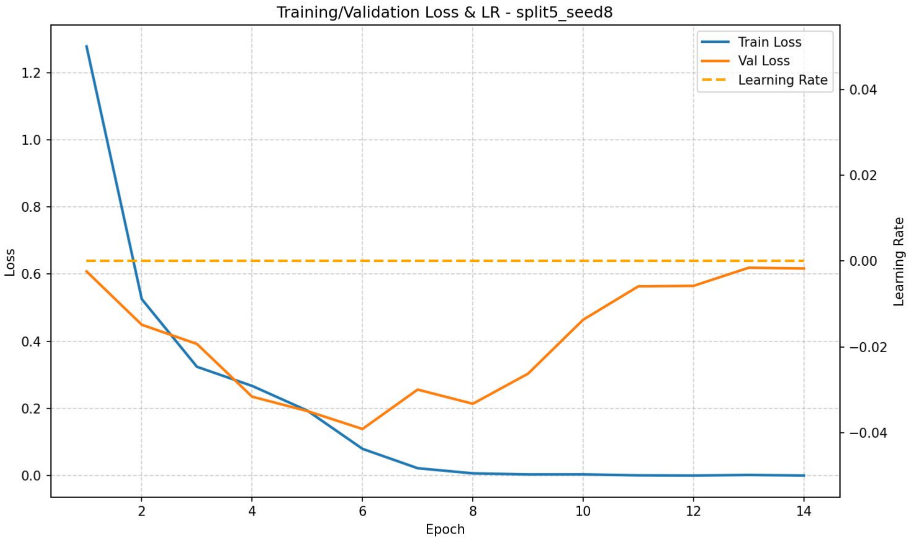
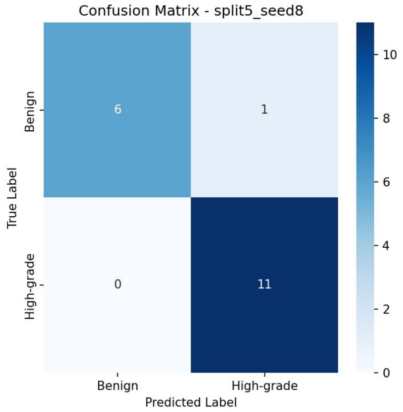
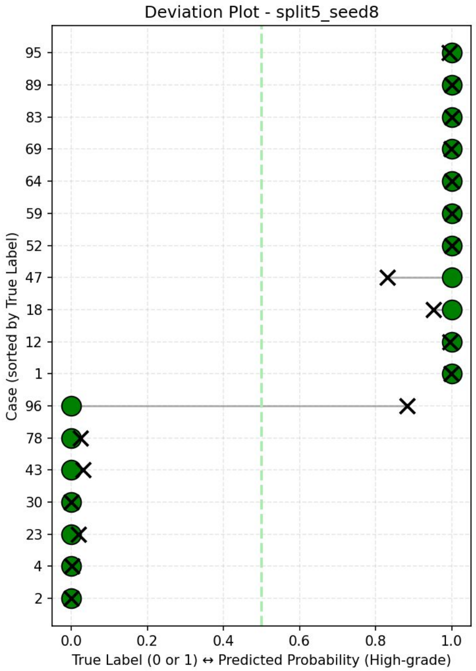
Max Split 5 Seed 8







Split 5 takeaways?

/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/visualization.ipynb
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/projects/e32998/STAT390_Winter2026/Presentation4/STAT390_Winter2026_Team2/Presentation4_Charlene_Visualization/Plots_split_seed

Visualization Takeaways

- Seed 24 is the minimum accuracy seed in Splits 2, 3, and 4, and shows the most volatile training curves — it seems to lead to particularly unstable optimization.
- Seed 34 underperforms in Splits 1 and 5.
  - These two seeds may be worth deprioritizing for future runs.

- Cases 56 (Split 1), 26 and 86 (Split 2), 46 (Split 3), 77 (Split 4), and 96 (Split 5) are the persistent false positives — benign cases the model almost always gets wrong regardless of seed within their respective splits. These are the highest-priority cases for biological review.

- Rising val loss (Split 3 Seed 68, Split 5 Seed 8) is a pattern worth monitoring — it may indicate the early stopping criterion is not catching degradation effectively.
- Split 4 appears most learnable (achieves perfect accuracy, lowest FP rate) while Split 2 appears hardest (highest val loss, most FPs).

Task Reflection

- **Seed seems to matter for ~1 in 4 cases:** 73.1% of cases are perfectly stable across seeds, but patch ordering affects classification outcomes for borderline cases more near the decision boundary.
  - Split composition sets overall difficulty; seed determines individual case outcomes within that.
- **The most difficult cases are seed-independent:** cases 56 and 46 are consistently wrong regardless of patch ordering, which suggests similar suboptimal solutions are being reached
  - Others like case 15 or 45 at 50% flip between correct and incorrect across seeds evidence of different local optima being reached depending on initialization
- **Implication for evaluation and tuning:** false positive counts are notably seed-sensitive (split 2 ranged from 1-4, Split 4 from 0-1), meaning a single seed run could give a misleading picture of model performance
  - → multiple seeds should be used for robust evaluation

Task Reflection

- Within a split, patch ordering (seed) can swing accuracy by up to 17 percentage points
  - This is a meaningful difference caused only by which patches the model sees first
- Early epoch patch ordering is influential
  - Seeds that produce clean val loss descent in epochs 1–5 consistently achieve better final outcomes
- Seed effects are split-dependent and non-transferable
  - Seed 24 is among the best in Split 1 but the worst in Splits 2, 3, and 4
  - No seed can be universally trusted across data compositions

Future Directions

- **Use multiple seeds for hyperparameter tuning:**
  - Given that seed can swing accuracy by up to 17 percentage points within a split, single seed and split Optuna trials risk selecting hyperparams that only perform well under a lucky initialization
- **To address persistent false positives:**
  - Cases 56, 46, 86, 77, and 96 are consistently wrong across seeds via visualizations. Biological review of these cases could inform targeted reweighting strategies
- **Cross reference high attention patches across seeds:**
  - Attention outputs saved for all 50 runs, so given results it is worth examining whether borderline cases attend to the same patches regardless of seed, distinguishing consistent misleading features from unstable optimization

Project Reflection

Key Learnings

- Learned how to navigate GitHub and shared repos
- Reinforced idea from STAT 303 sequence that model evaluation cannot stop at accuracy - many metric and visualizations are necessary explore!
- Learned core concepts of MIL, deep learning architecture, training workflow
    - Bag-level, instance-level
    - New hyperparameters such as epochs, batch size, dropout
- Gained experience using VS Code, terminal commands, and debugging
- How to structure collaboration across distinct roles on a shared question

What we would do differently

- Have a mathlete
- All team members should understand the code from the beginning, not only the coders
  - Coders and analysts together from the beginning - go hand in hand
- Focusing on one area of improvement or concept implementation for the model
- Either separate role-based groups who hand off work to each other to work collaboratively on or coordinate more within the mixed-function groups
  - Different base levels of knowledge - could help to collaborate with others within same function of more expertise

Real-world data science applications

- Interpretability and error analysis are necessary
- Working in a multidisciplinary team where technical and domain expertise must be actively integrated
- Importantly leveraging AI to assist with specific instructions for efficient coding, and which tasks AI struggles with where human analysis is necessary
- Working and becoming comfortable with existing code and architecture
- Importance of reproducibility and experiment tracking to ensure results can be replicated and trusted

Part 2: Biological Analysis

Adithi & Jason

Task Breakdown

1. Identify features the model may not yet be identifying well
    a. Identify patterns in the features that the model appears to detect successfully
    b. Identify features that could potentially confuse the model, but where the model may have become sufficiently robust to avoid misclassification
2. Determine if there is arbitrary or lack of a clear systematic pattern in feature identification
3. Propose future directions based on findings
4. Reflection

Case 86: Common Misclassified Benign Case

Task 1

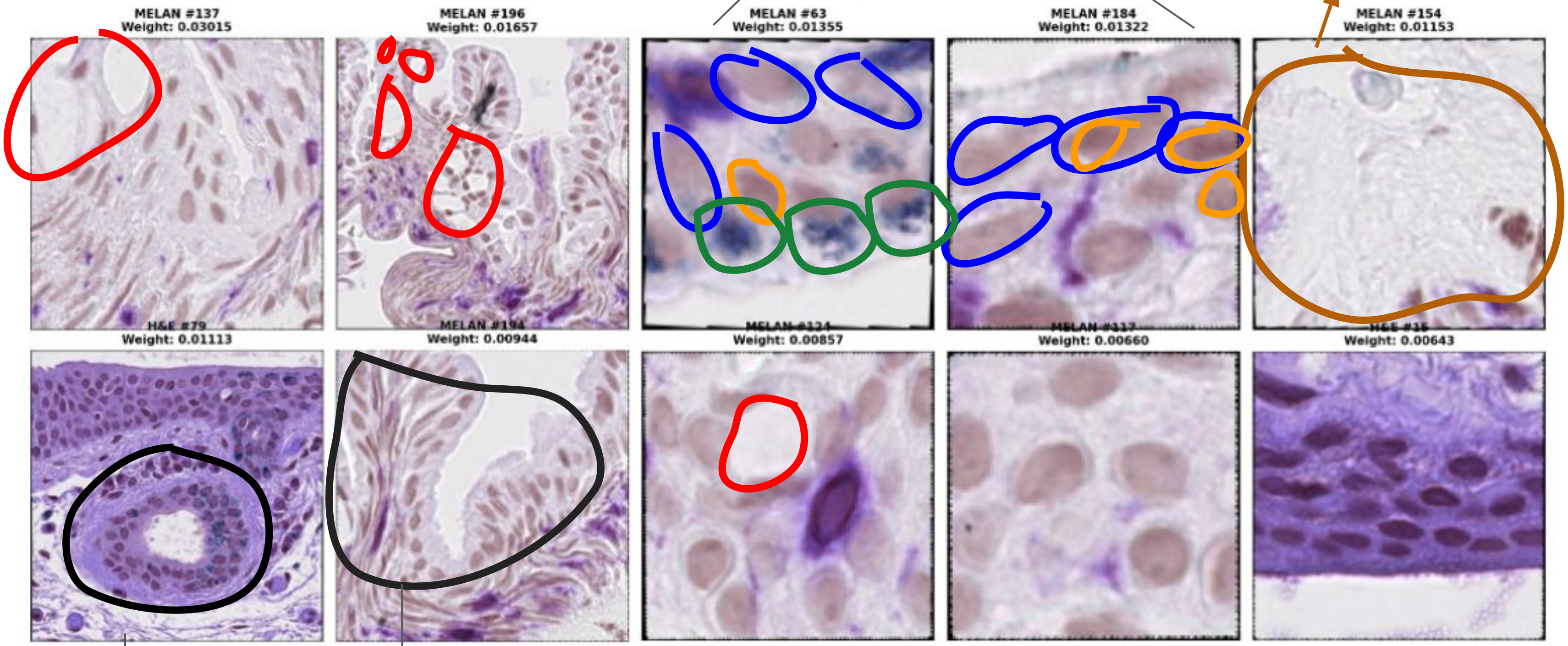
Team 1 - Case 86 (Benign, Misclassified)

Goblet cell hyperplasia

Slight melanosis

Some elongated cells, nuclear hyperchromasia

Patch appears to be of stroma



Invagination

Could be interpreted as invading, but is likely an outer fold of the slice

Normal cell size, shape, distribution

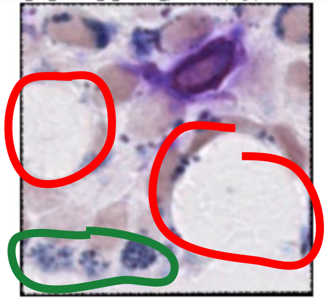
Team 2 - Case 86 (Benign, Misclassified)

<u>Patches deemed "most high-grade"</u>

Melanosis

case_86_match_2_melan_patch56.png | Prob: 1.000

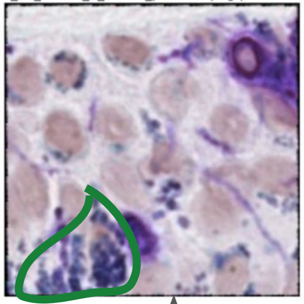
case_86_match_2_melan_patch23.png | Prob: 1.000



Goblet cell hyperplasia

case_86_match_1_melan_patch167.png | Prob: 1.000

case_86_match_2_melan_patch17.png | Prob: 1.000

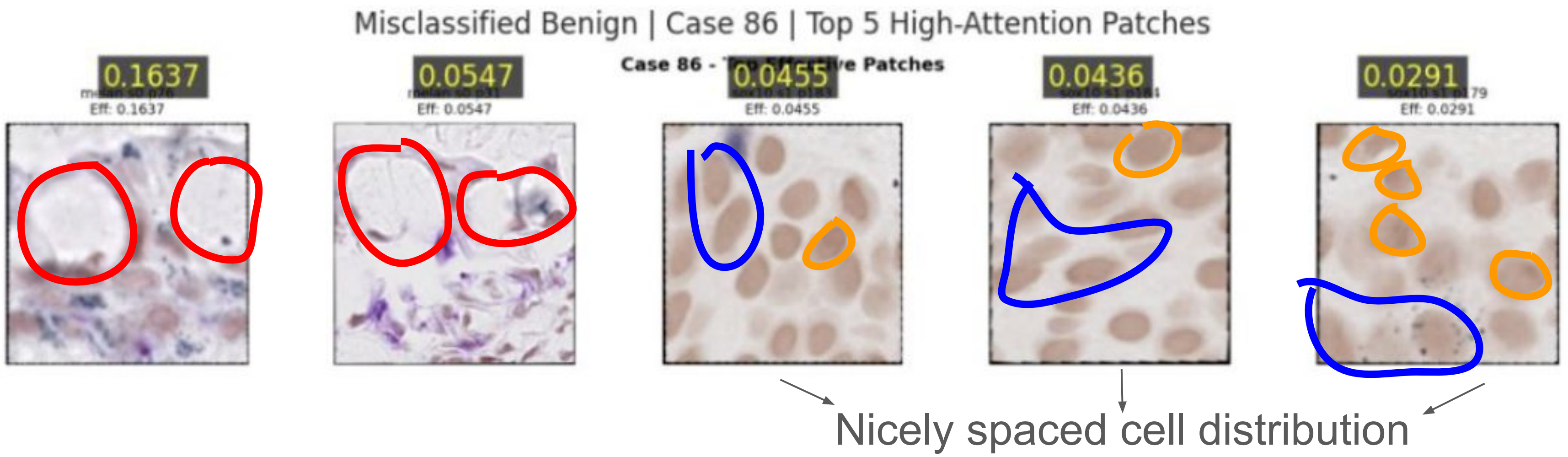


case_86_match_2_h&e_patch140.png | Prob: 1.000

Mostly typical cell distribution, shape, size

Team 2 - Case 86 (Benign, Misclassified)

Highest attention patches



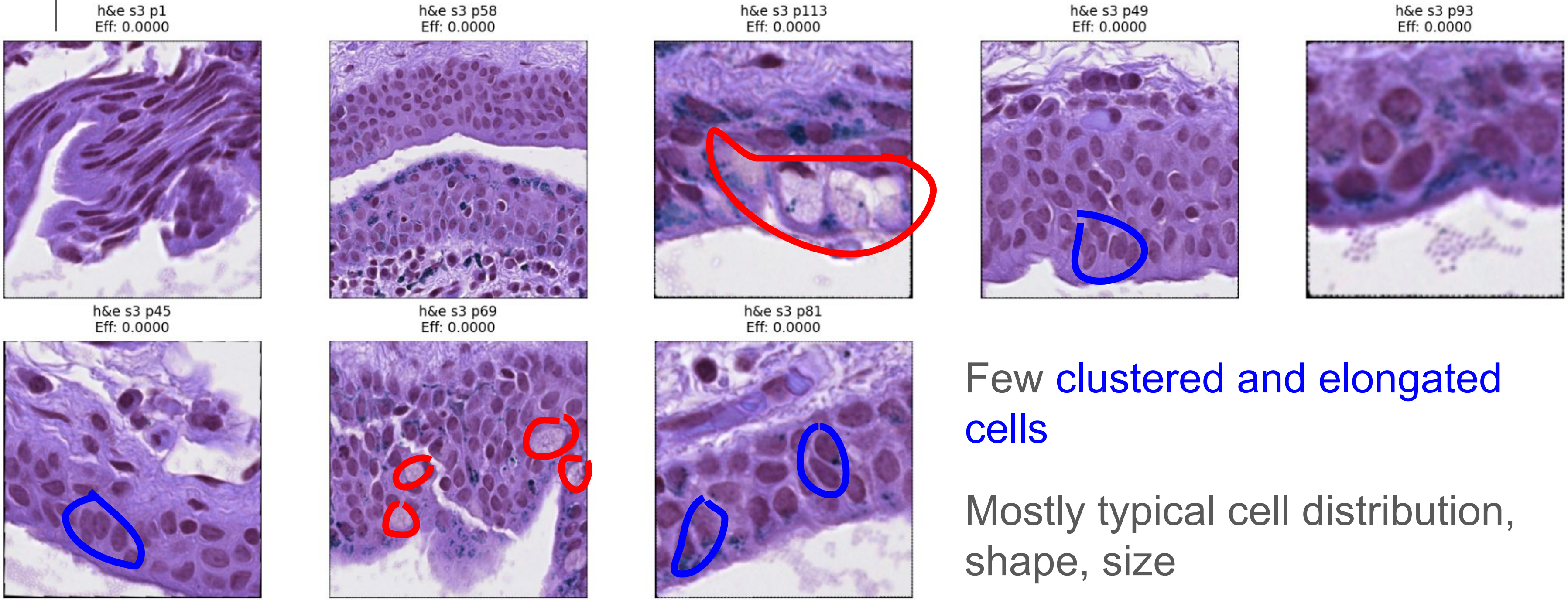
Goblet cell hyperplasia, some slightly larger & elongated cells, and some nuclear hyperchromasia

Team 2 - Case 86 (Benign, Misclassified)

Top bottom-attention patches

Likely slice folding of outer membrane area

Goblet cell hyperplasia



Slight goblet cell presence

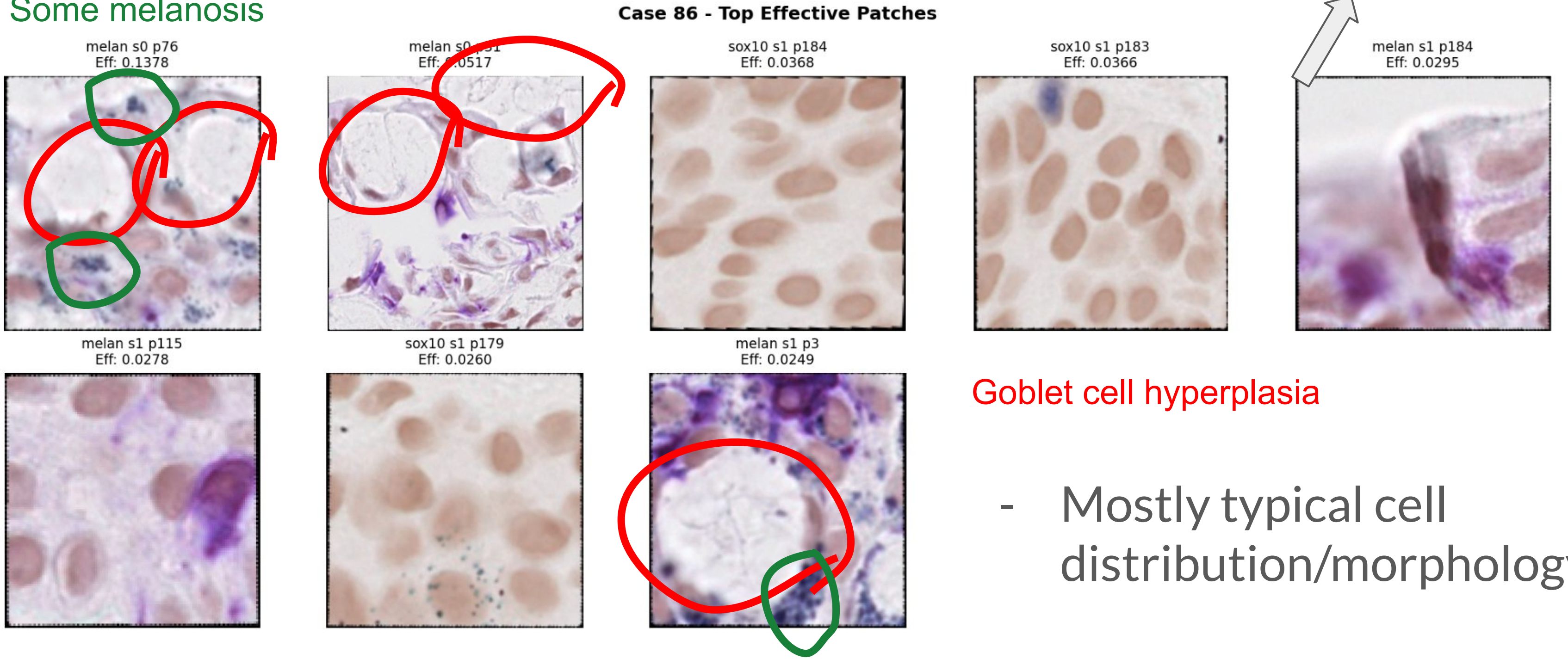
Few clustered and elongated cells

Mostly typical cell distribution, shape, size

Team 3 - Case 86 (Benign, Misclassified)

Some melanosis

Low resolution/artifact

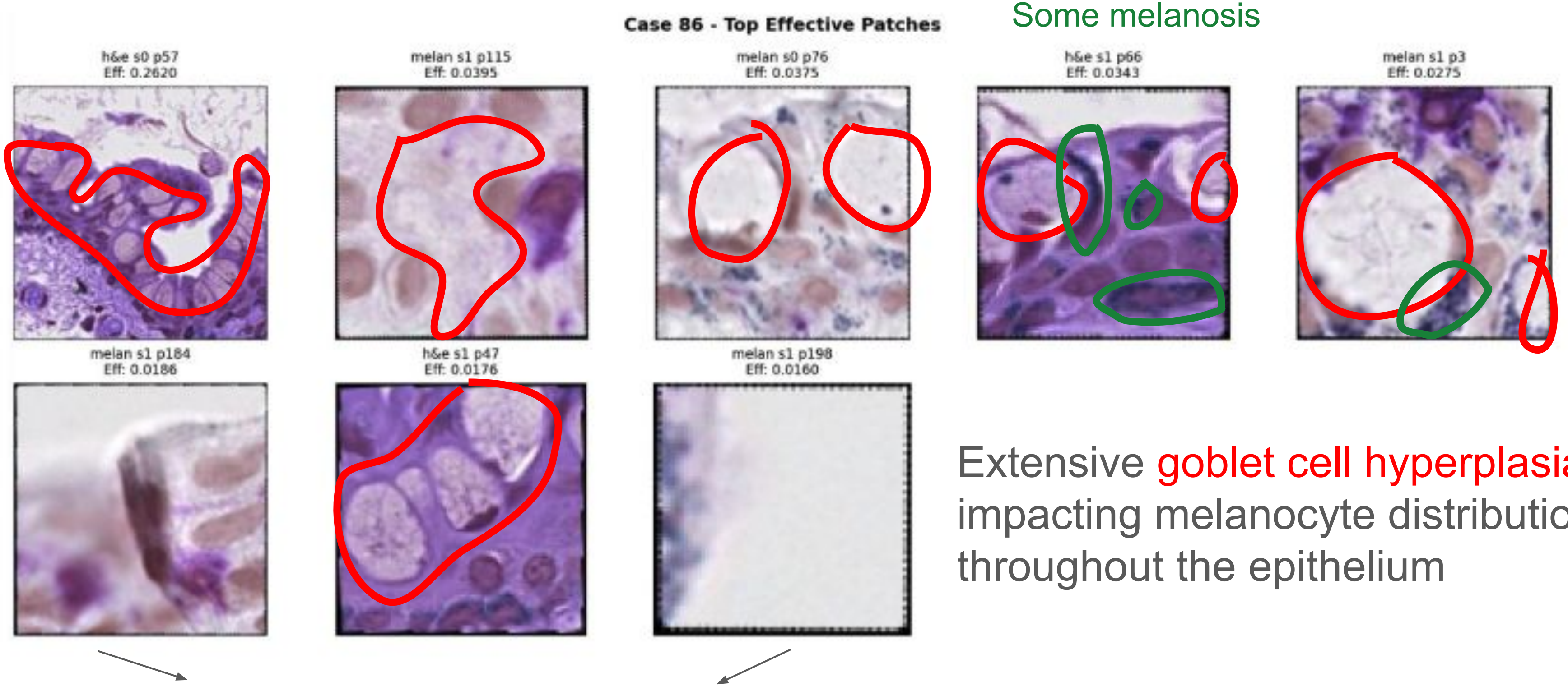


Goblet cell hyperplasia

- Mostly typical cell distribution/morphology

Team 4 - Case 86 (Benign, Misclassified)

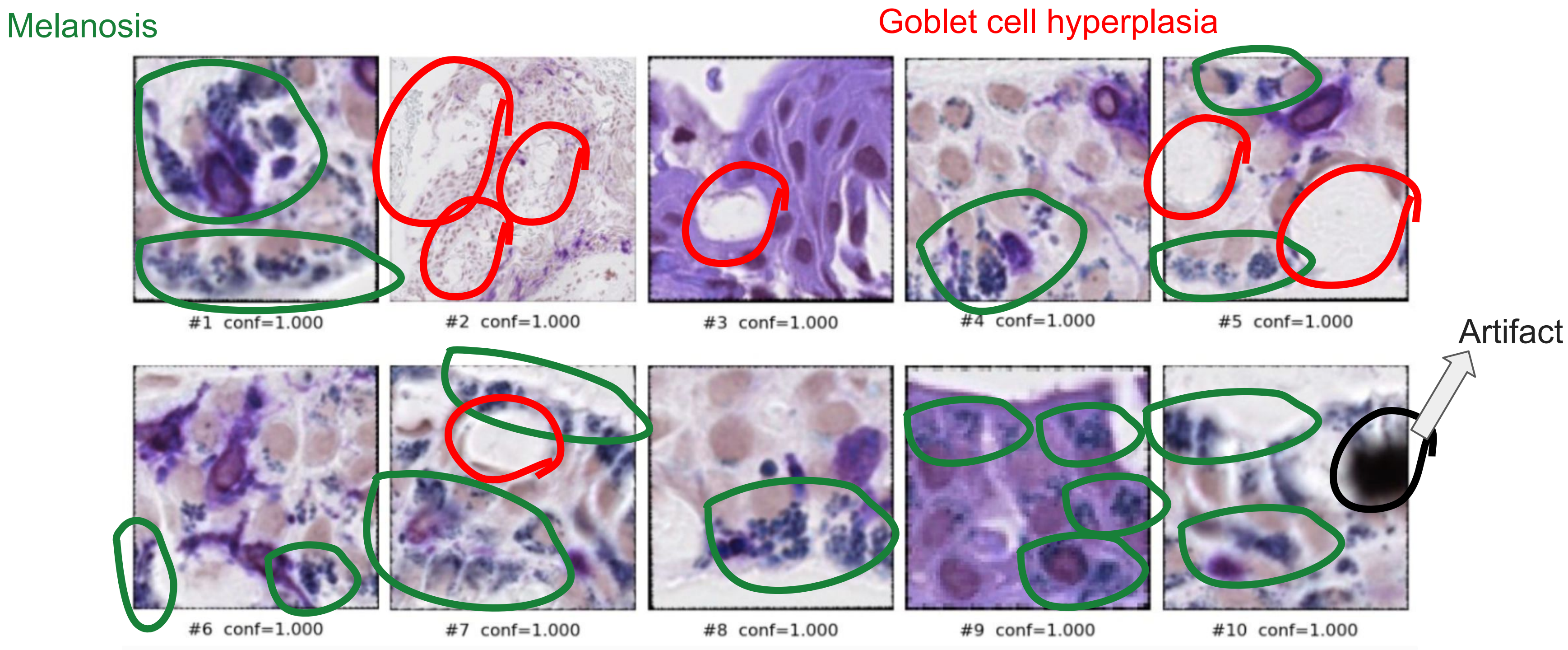
Some melanosis



Extensive goblet cell hyperplasia impacting melanocyte distribution throughout the epithelium

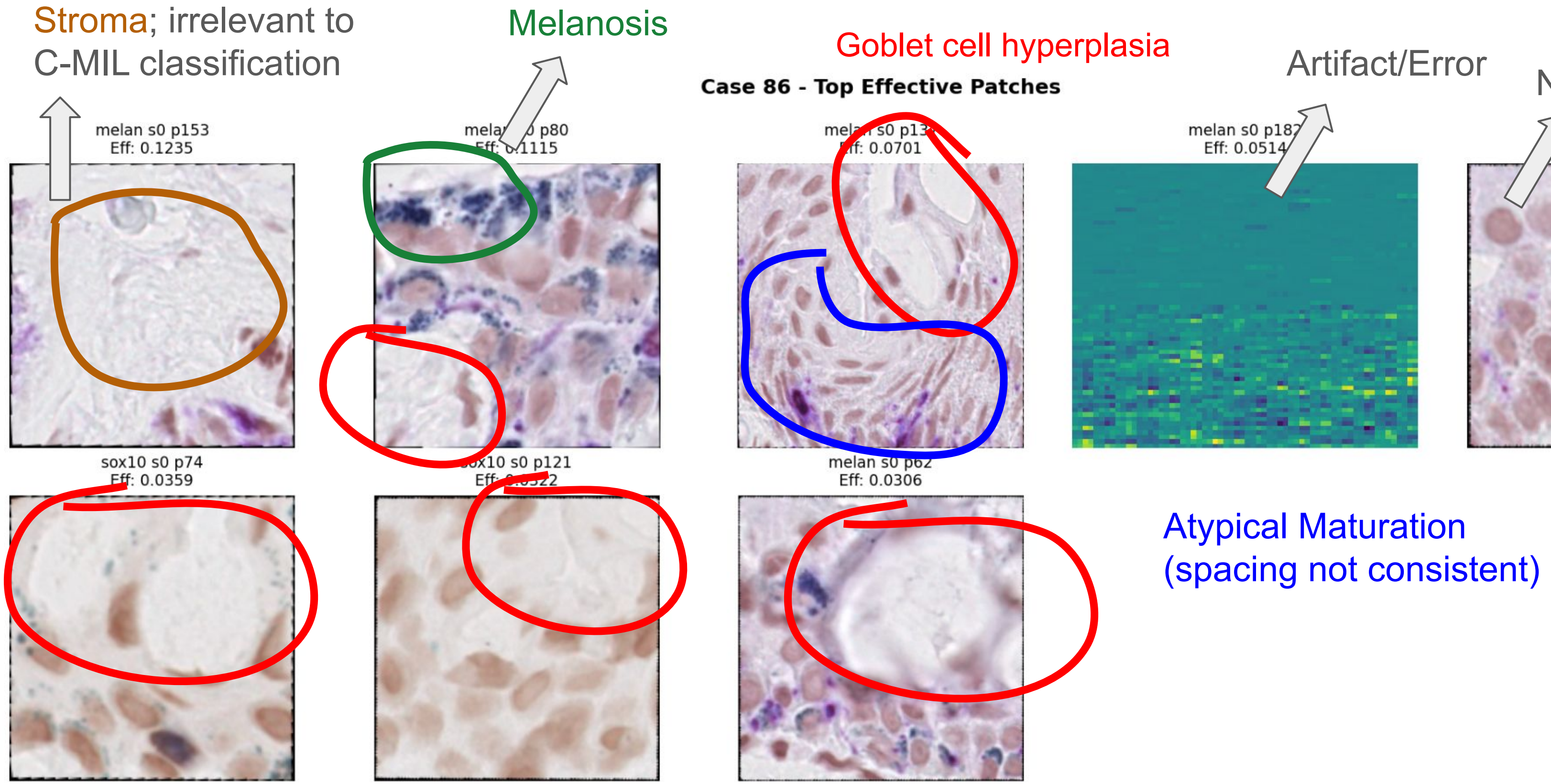
Artifacts: classification should not be based on these non-biological capturings

Team 5 - Case 86 (Benign, Misclassified)

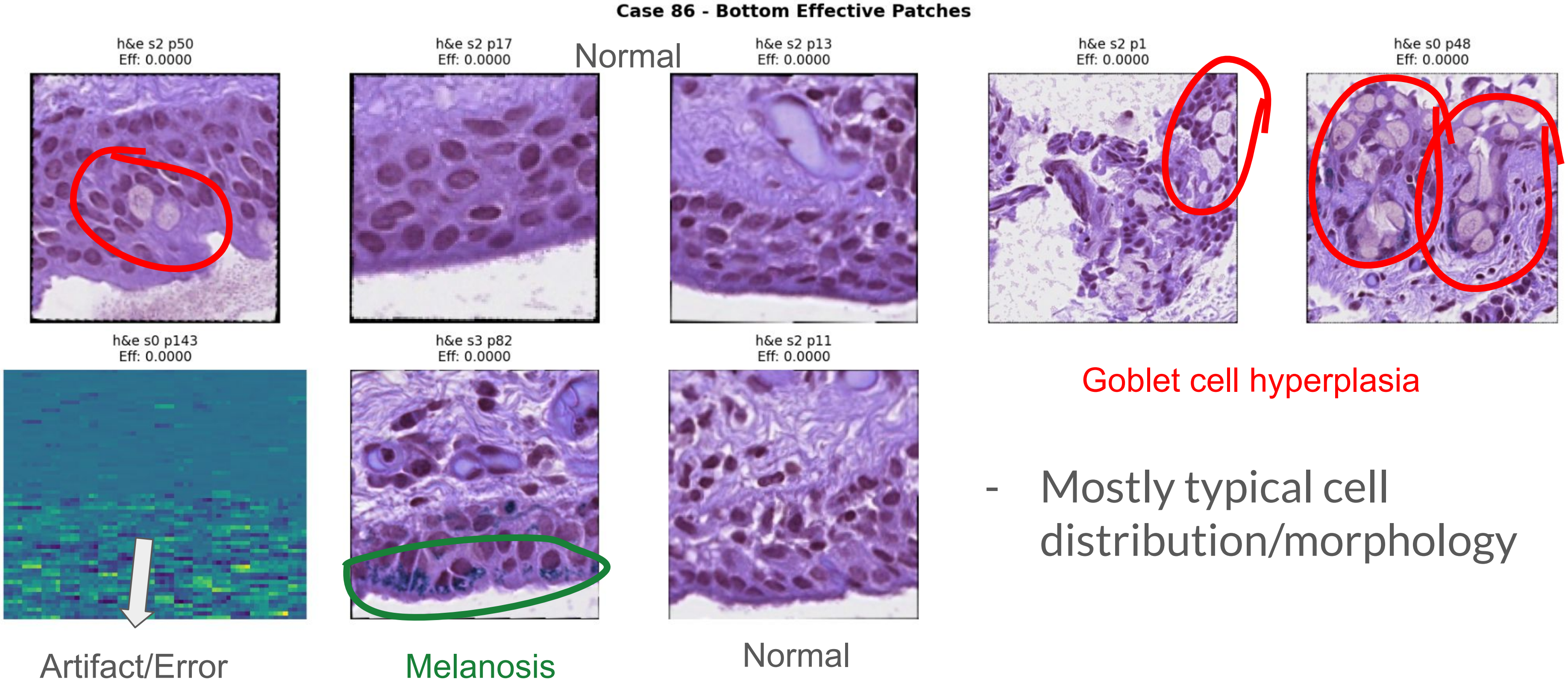


- Mostly typical cell distribution/morphology

Team 6 - Case 86 (Benign, Accurately Classified)



Team 6 - Case 86 (Benign, Accurately Classified)



- Mostly typical cell distribution/morphology

**Bottom effective patches

Takeaways

- Heavy goblet cell hyperplasia noticed across high-attention patches for each team
  - This is a feature we noted in Presentation 2 could contribute to misclassification of benign cases due to impacts on melanocyte distribution
- Melanosis and some atypical maturation was also observed across many teams' high attention patches
  - Cell elongation and nuclear hyperchromasia seen in some; not across all
- High attention patches lean mostly melan-A, but there is a good distribution across stains

** We noticed some of the high-attention patches (teams 1 & 6) were taken from the stroma, which is often irrelevant for C-MIL classification

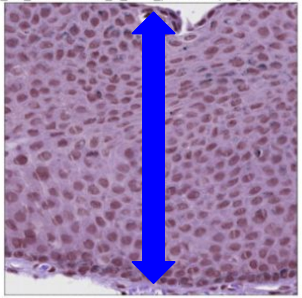
Case 21: Common Misclassified Benign Case

Task 1

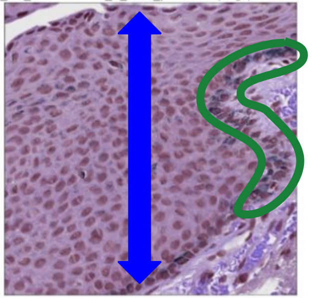
Team 2 - Case 21 (Benign, Misclassified)

Patches deemed "most high-grade"

case_21_unmatched_3_h&e_patch97.png | Prob: 1.000



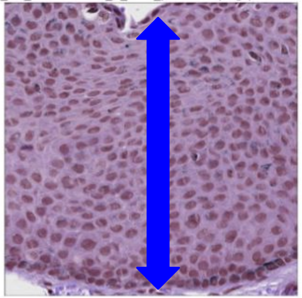
case_21_unmatched_3_h&e_patch94.png | Prob: 1.000



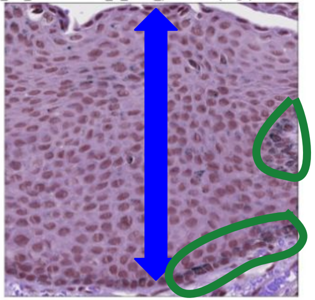
High cell density in epithelium, with few bigger or elongated cells

Some mild melanosis at basal layer

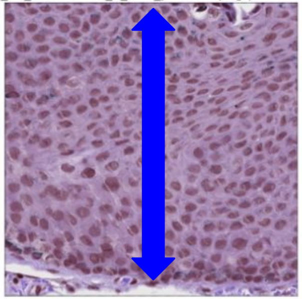
case_21_unmatched_3_h&e_patch96.png | Prob: 1.000



case_21_unmatched_3_h&e_patch95.png | Prob: 1.000



case_21_unmatched_3_h&e_patch98.png | Prob: 1.000



Team 2 - Case 21 (Benign, Misclassified)

Highest attention patches

Slight melanosis

Mostly normal cell distribution, shape, & size

Likely folded part of slice

Some larger cells & some clustering

Team 3 - Case 21 (Benign, Misclassified)

**Same top 4 patches as team 2

Mostly normal

Some melanosis

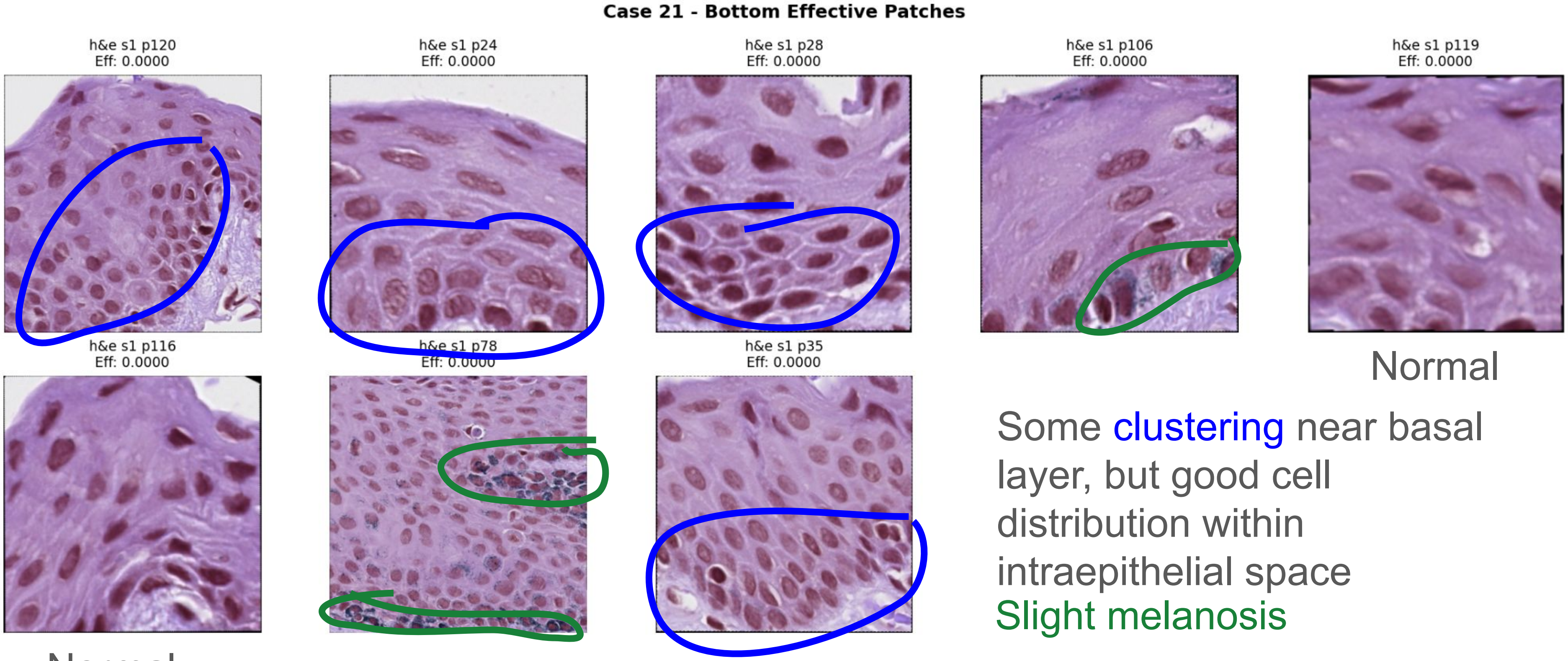
Some larger cells & some clustering

Uniform cell shapes & normal distribution, but slightly elongated

Some clustering at basal layer, but mostly normal

Likely folding – same as team 2

Team 3 - Case 21 (Benign, Misclassified)



Normal

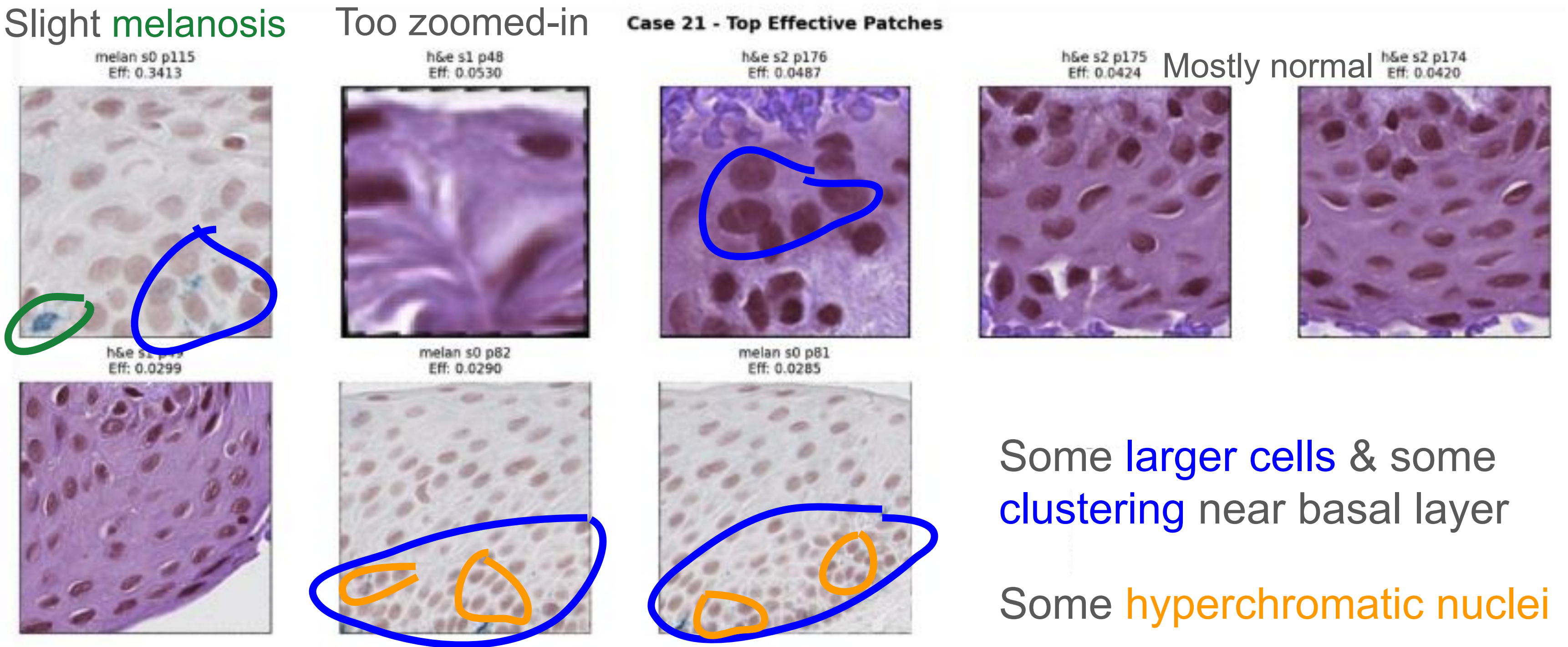
Normal

Some clustering near basal layer, but good cell distribution within intraepithelial space

Slight melanosis

**Bottom effective patches

Team 4 - Case 21 (Benign, Misclassified)



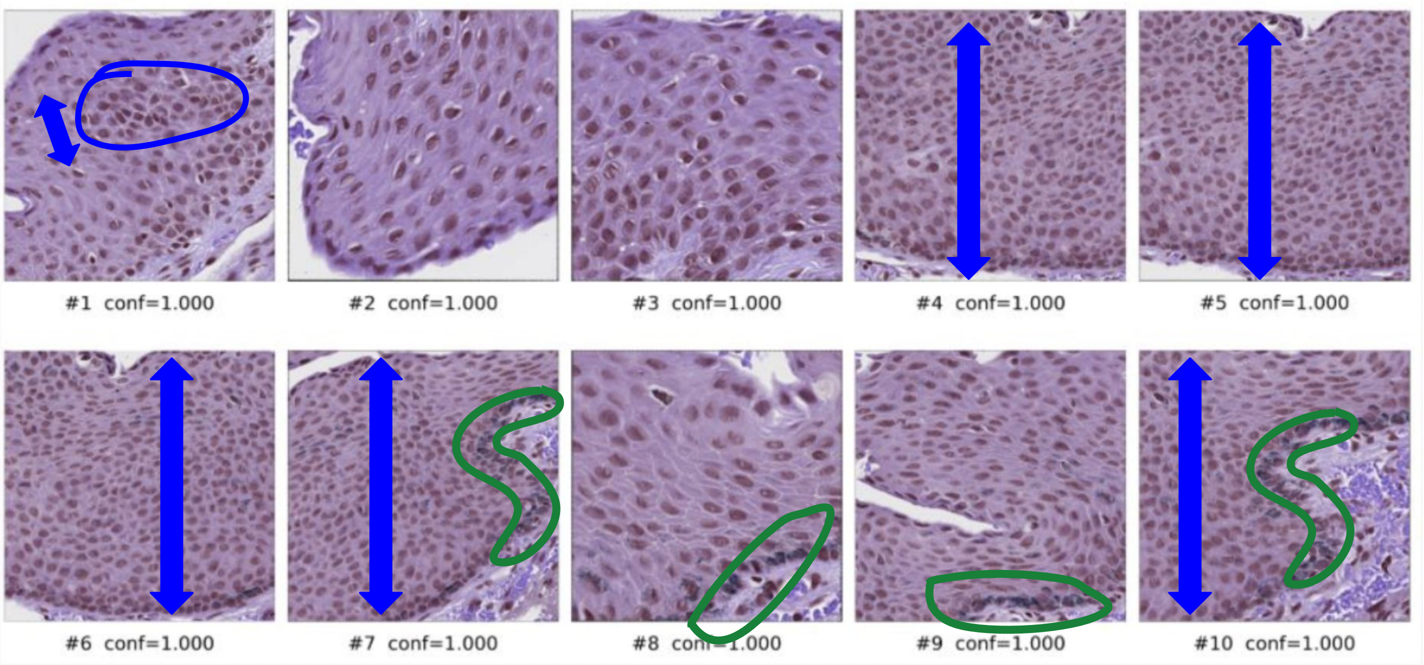
Normal

Some larger cells & some clustering near basal layer

Some hyperchromatic nuclei

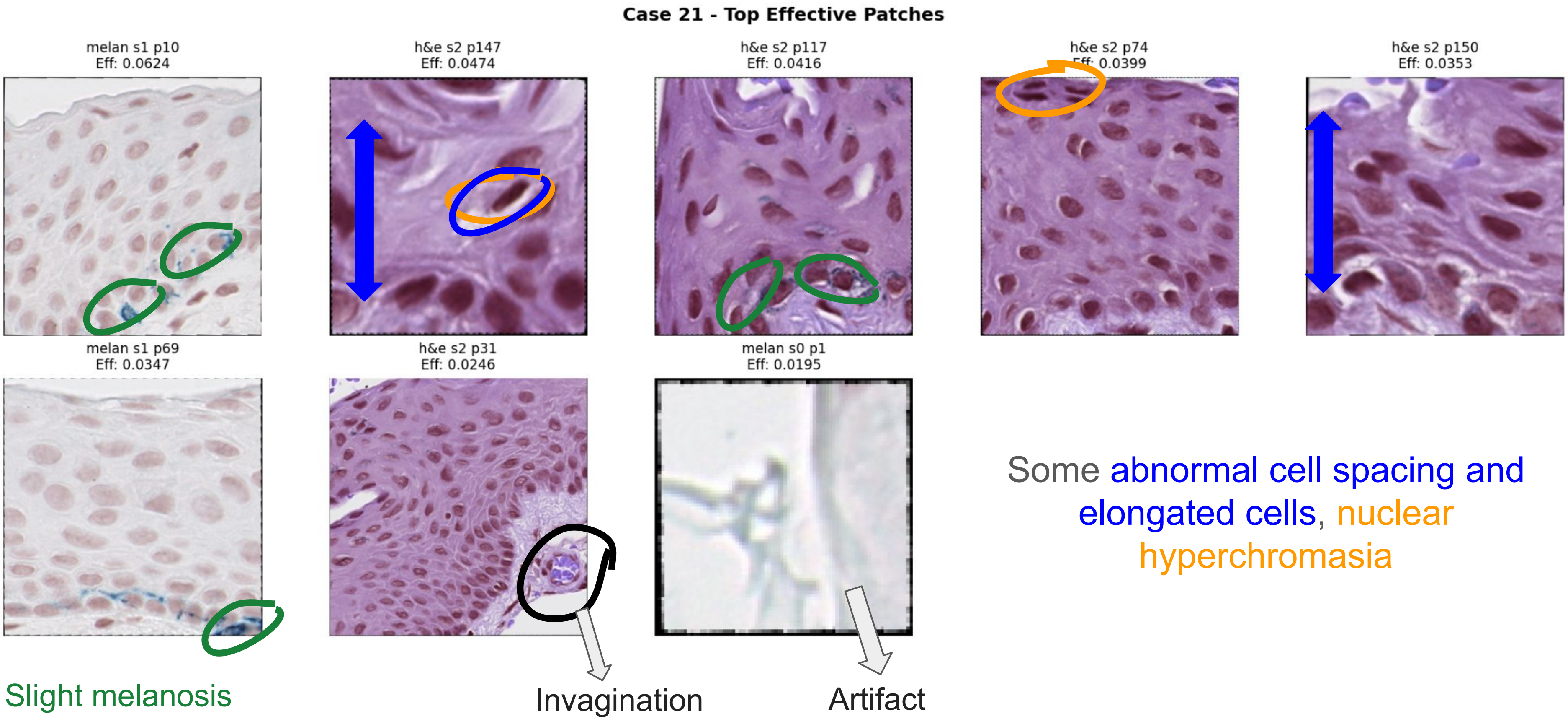
** Similar top patches to teams 2 & 3

Team 5 - Case 21 (Benign, Misclassified)



Some abnormal cell spacing, high cell density, consistent melanosis at basal layer

Team 6 - Case 21 (Benign, Misclassified)

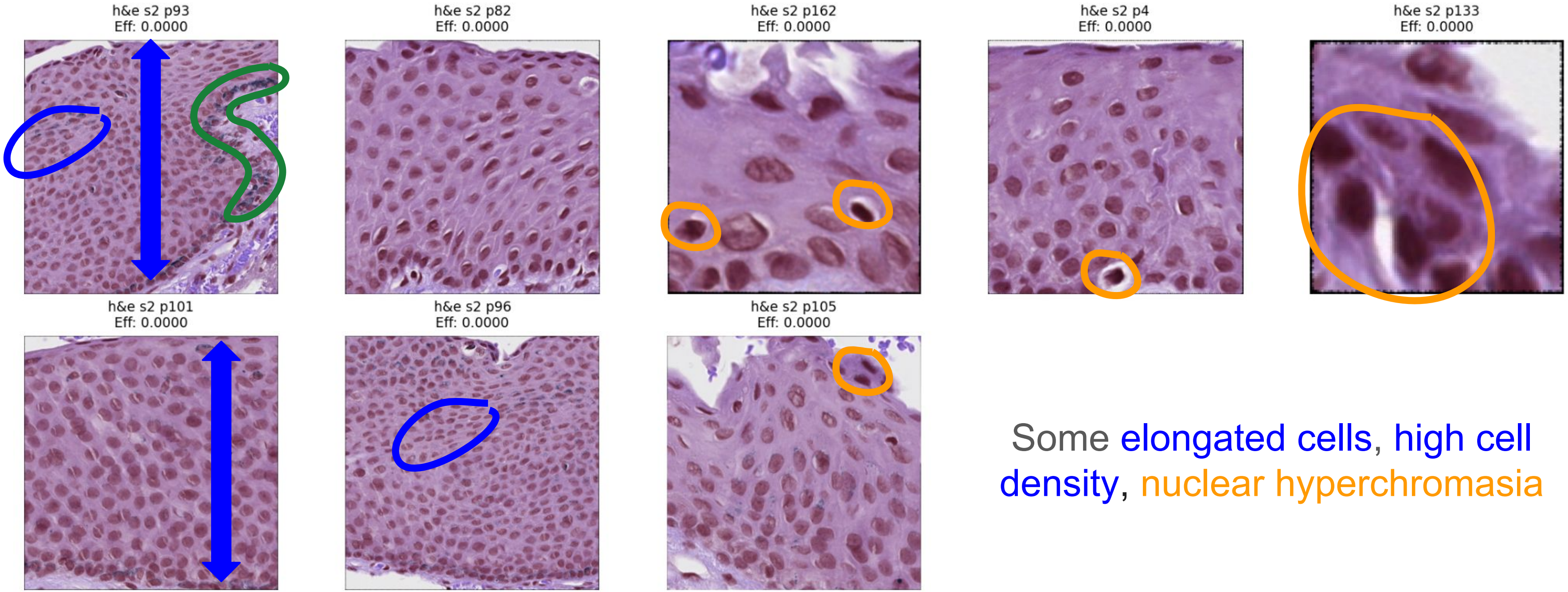


Some abnormal cell spacing and elongated cells, nuclear hyperchromasia

Team 6 - Case 21 (Benign, Misclassified)

Slight melanosis along basal layer

Case 21 - Bottom Effective Patches



Some elongated cells, high cell density, nuclear hyperchromasia

**Bottom effective patches

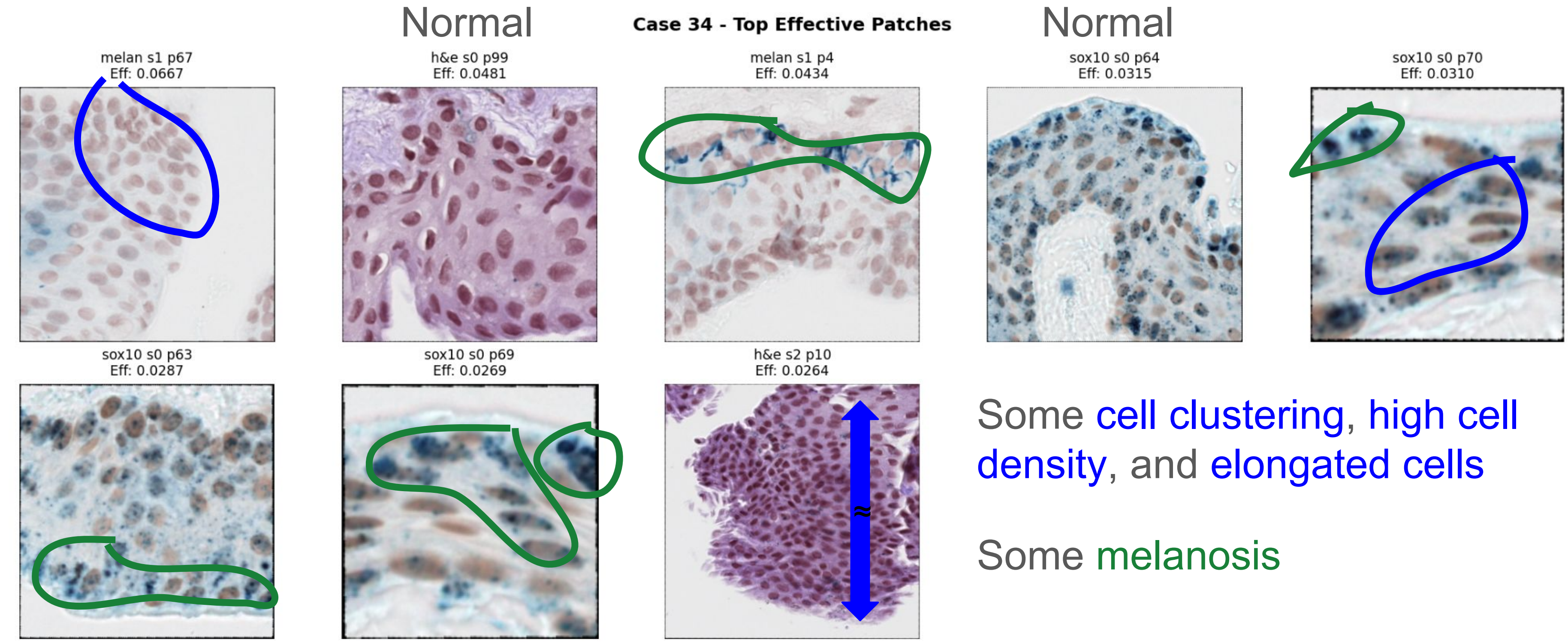
Takeaways

- A lot of the same high-attention patches are being used
  - Less variability than in case 86 across the different teams
- Not as many zoomed in patches
- High-attention patches are H&E heavy
  - Distinct from presentation 3 findings across teams in which Melan-A was the most important stain for assigning a classification across most cases
- Fewer goblet cells
- Some bottom effective patches (team 6) resemble high effective patches (teams 2, 5)
  - Lack of consistency for models

Other Misclassified Benign Cases

Task 1

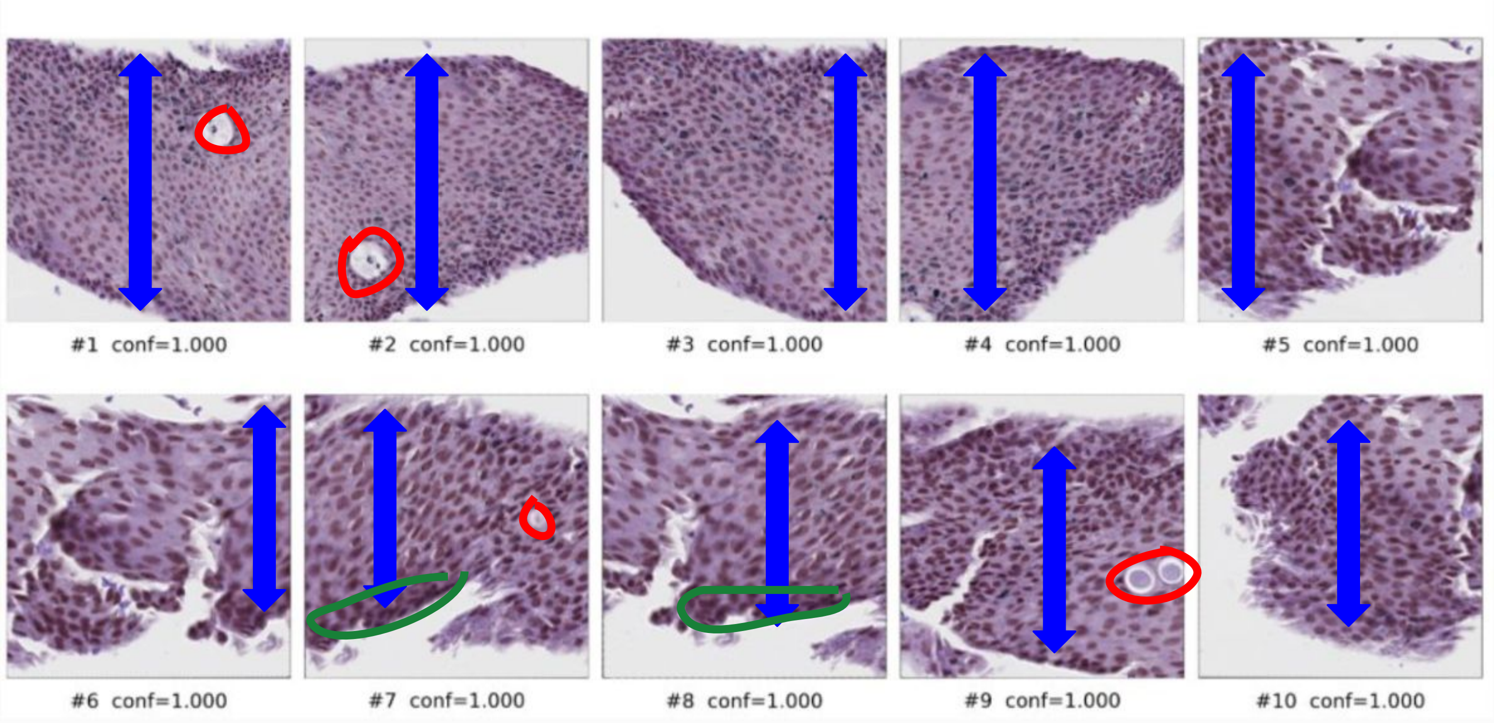
Team 3 - Case 34 (Benign, Misclassified)



Some cell clustering, high cell density, and elongated cells

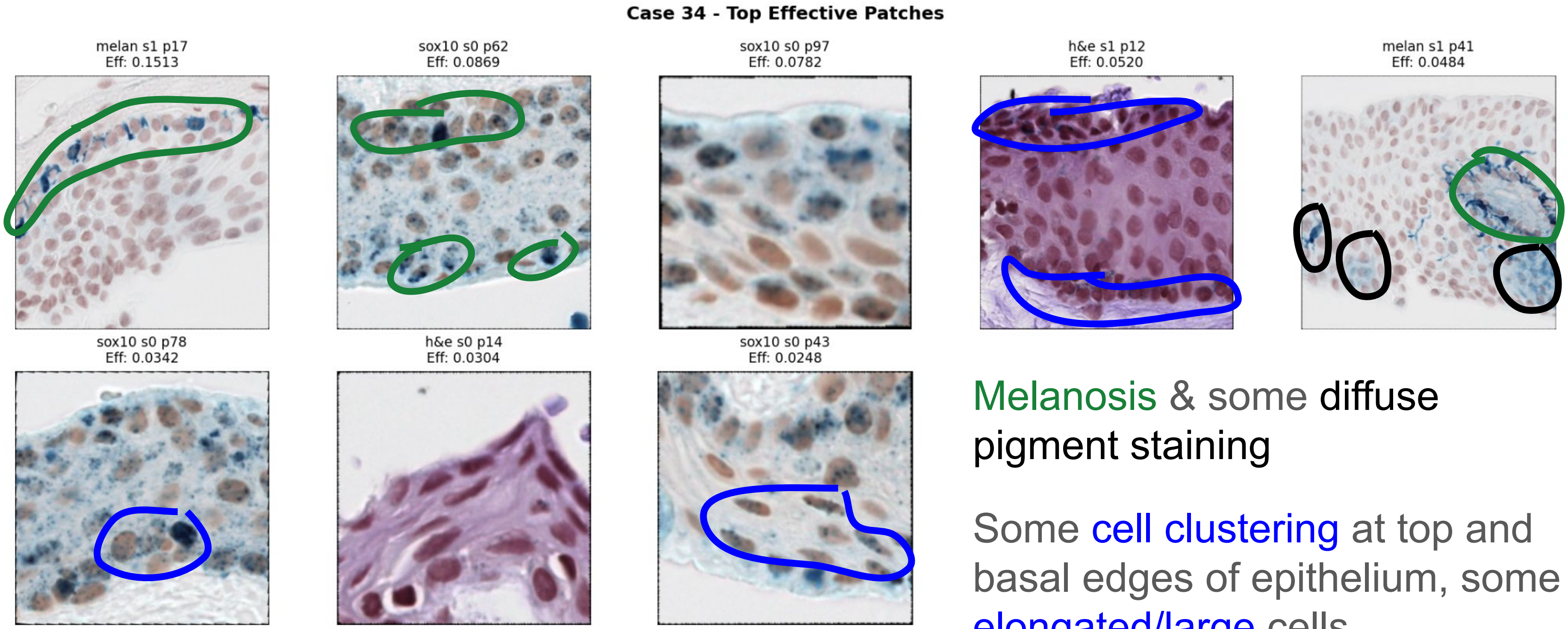
Some melanosis

Team 5 - Case 34 (Benign, Misclassified)



Some abnormal cell spacing and high cell density in all patches, some slight basal layer melanosis, few goblet cell clusters

Team 6 - Case 34 (Benign, Misclassified)



Outer edge tissue fold

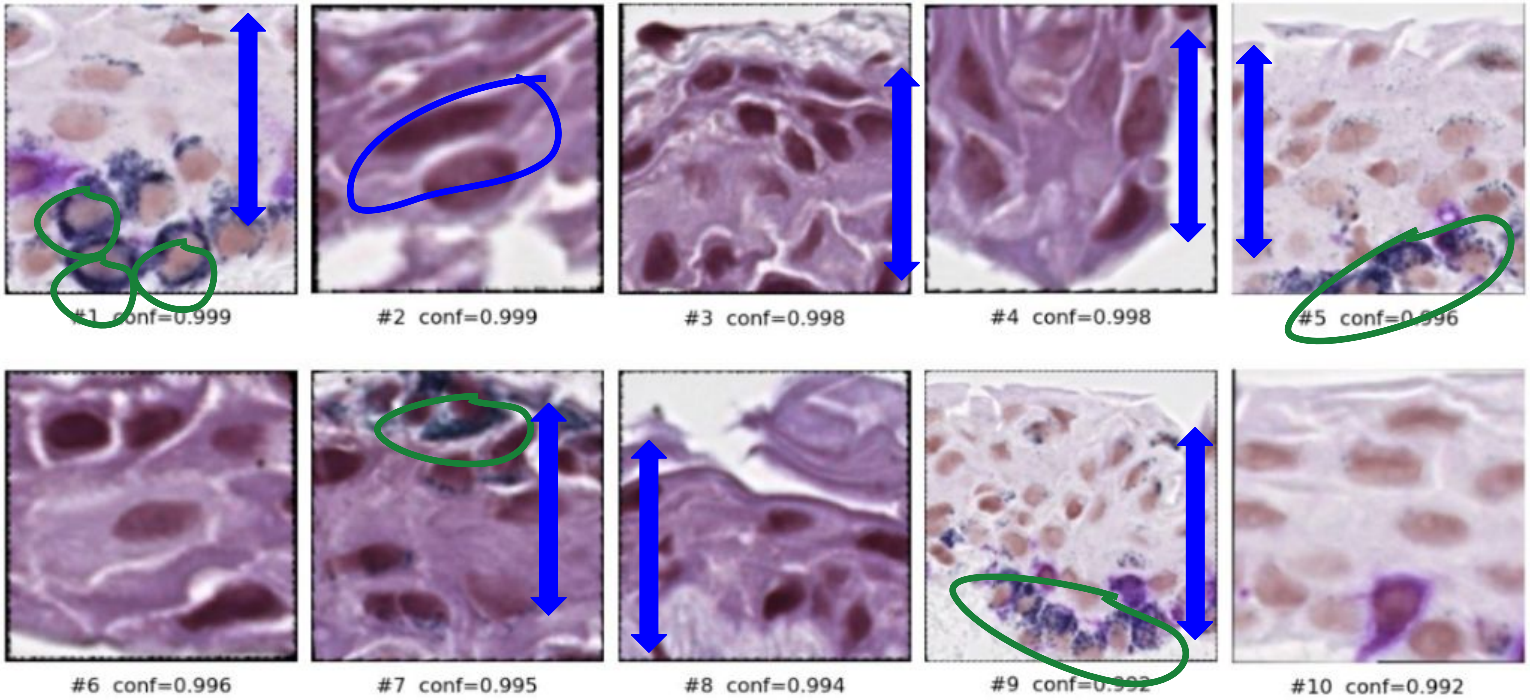
Melanosis & some diffuse pigment staining

Some cell clustering at top and basal edges of epithelium, some elongated/large cells

Team 2 - Case 96 (Benign, Misclassified)

Some atypical maturation and elongated cells

Team 5 - Case 96 (Benign, Misclassified)



Some cell clustering and elongated cells

Some melanosis

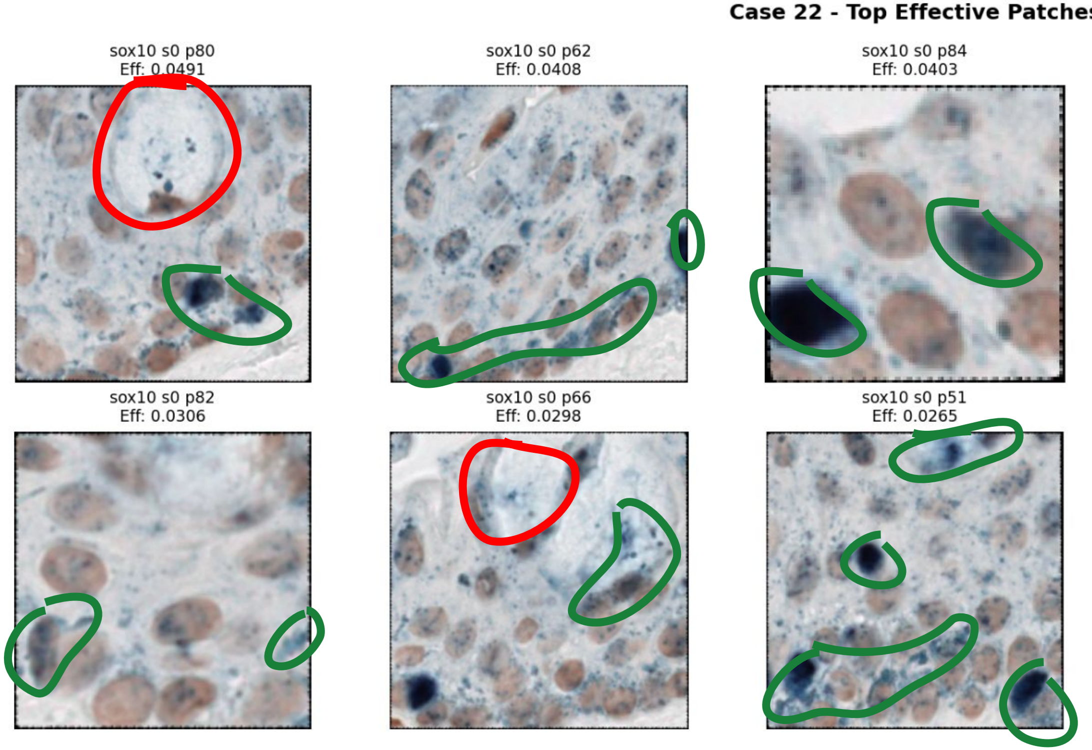
Takeaways

- Patches are fairly zoomed in
- Distribution of stains for high-attention patches is pretty equal besides team 5 case 34
- Less goblet cell hyperplasia
- Some diffuse melanosis across many patches
- Abnormal cell spacing, density, and shape in most of the high-attention patches, similar to case 21

Correctly Classified Benign Cases

Task 1

Team 3 - Case 22 (Benign, Hard to Classify, Properly Classified)

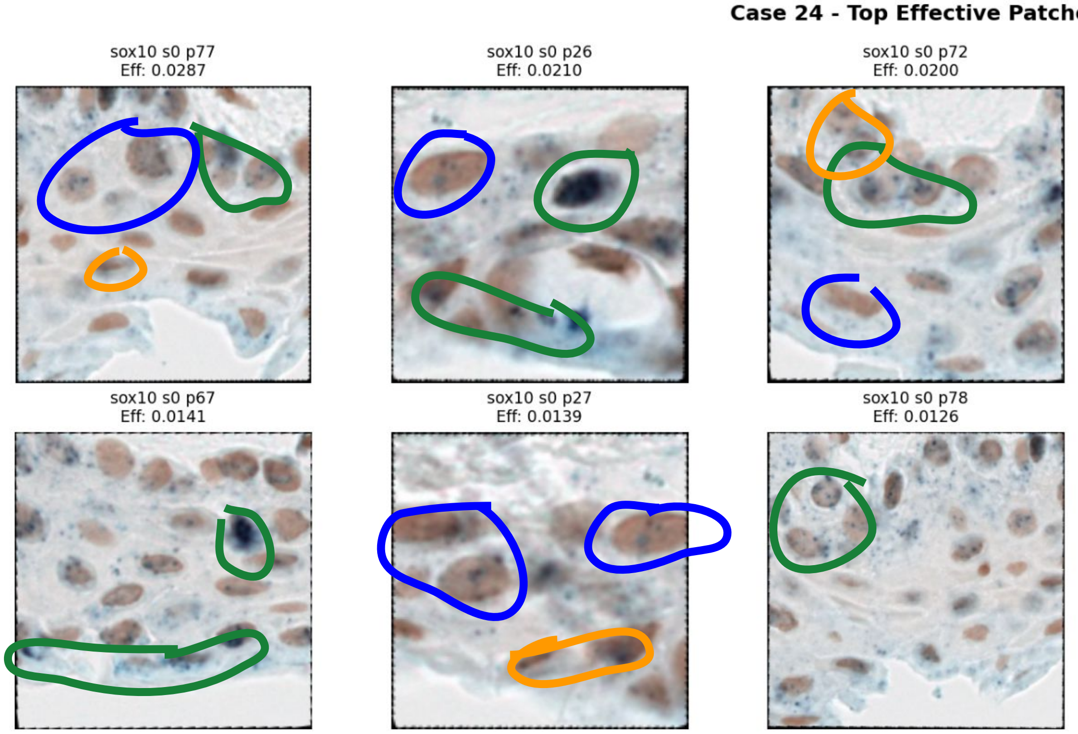


Good cell distribution, shape, size in all patches

Melanosis present

Some goblet cell presence

Team 3 - Case 24 (Benign, Hard to Classify, Properly Classified)



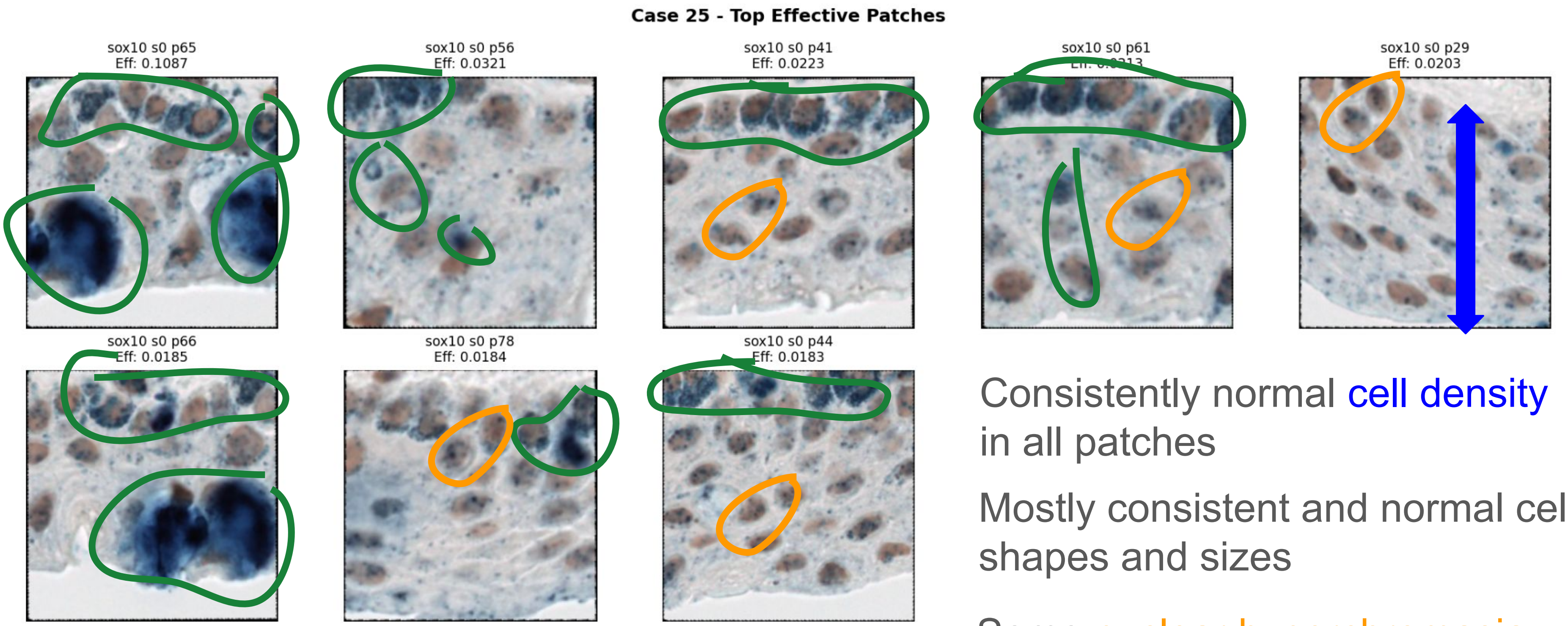
Consistently normal cell density in all patches

Some bigger & elongated cells

Melanosis present

Some hyperchromatic nuclei

Team 3 - Case 25 (Benign, Hard to Classify, Properly Classified)



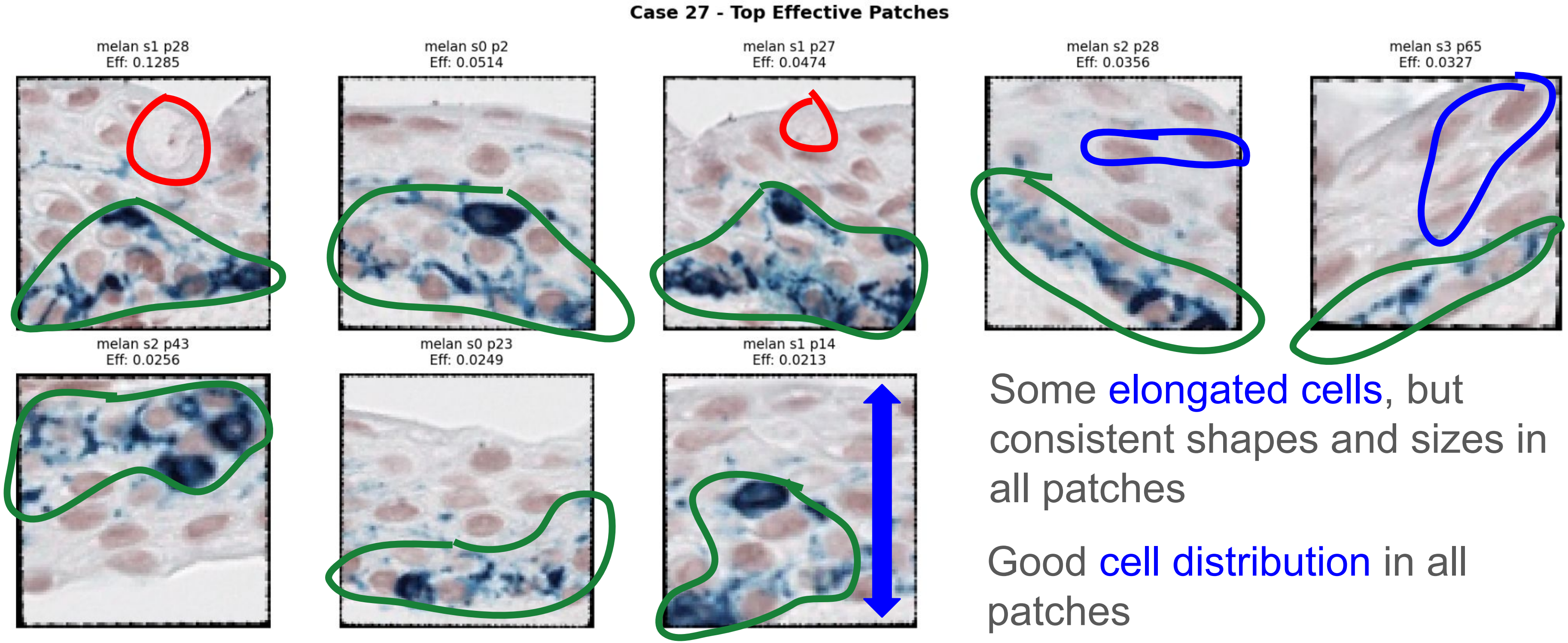
Consistently normal cell density in all patches

Mostly consistent and normal cell shapes and sizes

Some nuclear hyperchromasia

Significant melanosis

Team 3 - Case 27 (Benign, Medium Difficulty, Properly Classified)



Some elongated cells, but consistent shapes and sizes in all patches

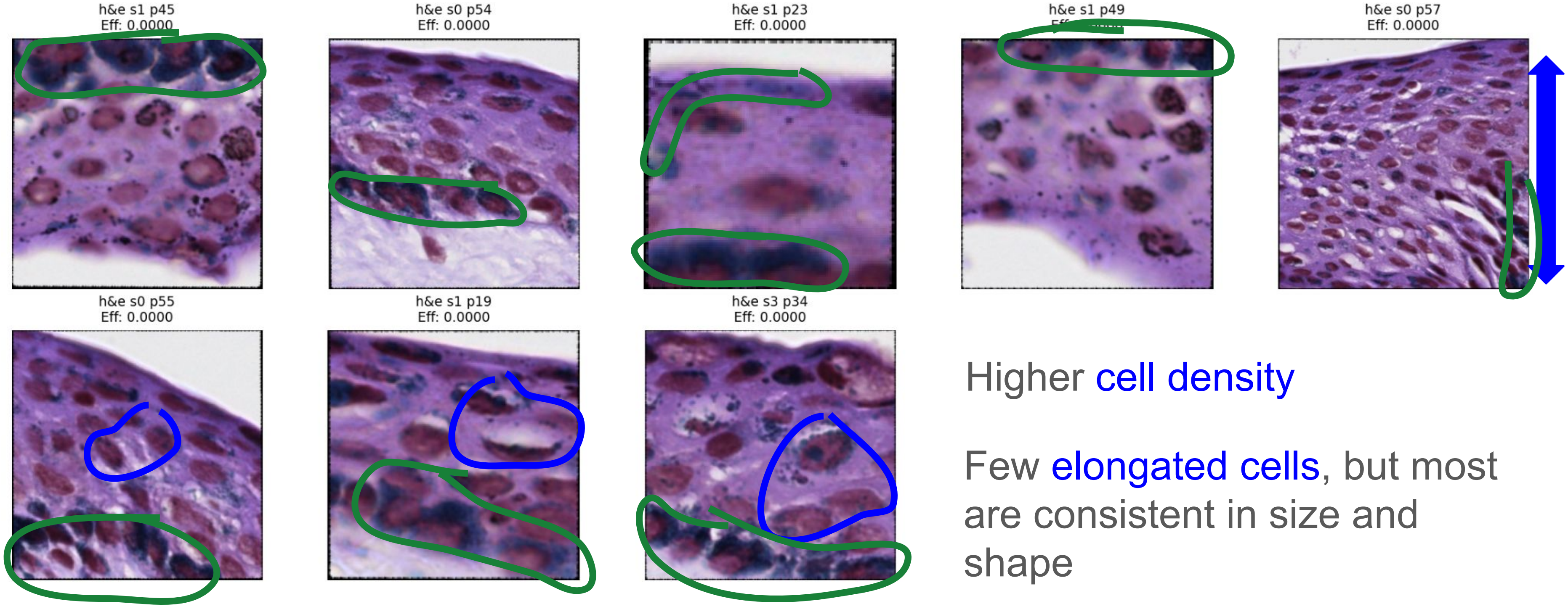
Good cell distribution in all patches

Significant melanosis

Very slight goblet cell presence

Team 3 - Case 27 (Benign, Medium Difficulty, Properly Classified)

Too zoomed-in



Higher cell density

Few elongated cells, but most are consistent in size and shape

Significant melanosis

**Bottom effective patches

Takeaways

- More zoomed in patches
- High-attention patches are entirely SOX10 and Melan-A heavy
    - Bottom-effective patches are H&E heavy
    - Sox10 and Melan-A are more specific staining for nucleus and melanocytes -> there's more contrast in their images
        - Perhaps giving them more attention weight corresponds to more accurate classifications by the model for hard-to-classify cases given the higher proportion of them seen here than in the high-attention patches for the misclassified examples?
- Fewer goblet cells
- A lot of melanosis
    - Many cases have melanosis clusters, unlike other misclassified cases that have melanosis more spread out along basal membrane (case 21)

Takeaways

Task 2

Overarching Takeaways

- Many patches are zoomed in and don’t span the full intraepithelial space or are of the stroma
  - This makes it difficult to see where certain features are located/contained (near basal layer, toward top of epithelium, spread throughout, etc.)
    - This is vital as C-MIL is specific to the epithelium, and certain features are considered abnormal/concern for C-MIL based on location and spread
- Majority of high attention patches for misclassified cases were from H&E and Melan-A stained slices, and most of the concerning features within such patches pertained to:
  - 1. cell distribution/shape/size/density
  - 2. goblet cell hyperplasia
  - 3. Melanosis

Cell distribution seems to be the main feature the model is not classifying well - properly classified benign cases had low cell density, but higher cell density cases with uneven distribution were consistently misclassified

Overarching Takeaways Contd.

- Perhaps the model has improved with regard to melanosis not being a sure-fire sign of high grade C-MIL
  - All properly classified benign cases have significant melanosis present
    - In the misclassified cases, melanosis is still present but to a lesser extent, so misclassification is likely due to the other aforementioned features (e.g. cell distribution, goblet cell hyperplasia)
  - Distribution of melanosis in misclassified cases is also more along basal membrane

Future Directions

Task 3

Future Directions

- Accounting for patch location relative to each full slice would allow for region & “extent” based analyses
  - As mentioned, region & spread of certain features can dictate how “concerning” they may be
  - For true high-grade cases, this modification would also help validate/verify that there are concerning features throughout the slice, improving classification certainty
- Find a way to ensure the training dataset is representative of the various features that a benign case can have, as that can impact the ability of the model to recognize said features in hard-to-classify cases

Project Reflection

Mandatory Task (#4)

Key Learnings From the Course

Technical Learnings

1. Model performance varies greatly depending on train/validation splits, underscoring the importance of representative sampling
2. Recognizing overfitting vs. generalization, especially with smaller biomedical datasets
3. A limitation of machine learning models is that they may learn from visual proxies (e.g. pigment contrast) instead of actual biological mechanisms
4. Importance of analyzing misclassified cases and attention maps, not just overall accuracy before modifying model parameters to enhance performance

Non-technical Learnings

1. Developed deeper understanding of the histology & pathology of C-MIL
2. Learned how biological features (e.g. melanocyte distribution, architecture, staining patterns) relate to model predictions
3. Practiced translating pathology observations into interpretable machine learning insights
4. Assigning tasks according to differential expertise to complete shared objectives

Considerations if Undertaking a Similar Project Again

1. Engage directly with project stakeholders (e.g. medical experts) earlier to refine methodology and evaluation and hear from their expertise
2. Use our biological and coding background to bridge communication between coders and medical expert stakeholders, identifying actionable implementations of medical expert ideas earlier in the process
3. Incorporate cross-stain reasoning earlier when designing model pipelines
    a. H&E is typically used first by pathologists to evaluate tissue architecture, and immunohistochemical stains like Melan-A and SOX10 are typically used afterward to confirm melanocyte distribution

Applying Learnings in Real-World Data Science

1. Combining machine learning techniques with domain expertise (e.g. medical, histological, pathological, biological)
2. Evaluating models using interpretability and error analysis, not just accuracy metrics, to understand why predictions are made
3. Conducting biomedical literature review to analyze the assumptions underlying the machine learning model
4. Learning how to collaborate and allocate tasks according to expertise within a multidisciplinary team to achieve a common goal

Takeaway: As future medical students and researchers, we can apply these processes and principles when evaluating clinical ML tools, ensuring models align with pathology, disease mechanisms, and clinical reasoning and can be used to tangibly improve health outcomes

References

GenAI tools

Fall 2025 Presentation 4

Winter 2026 Presentation 6

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